MoE Routing and Adaptation: Comprehensive Project Report

Recovery, data engineering, model adaptation, numerical export, evaluation, serving, and release

Report date: October 22, 2026 · Author/project owner: Ramanathan Swaminathan. This report consolidates verified project records and source inspection. AI-assisted analysis and existing AI answer assessments are identified as such. It is not an independent human evaluation, a claim of production readiness, or a claim that pretrained foundation models and recovered adapters were trained from scratch.

Current decision: experiments are complete and frozen; Stage 3 is not promoted. Answer-quality acceptance and production acceptance remain **false**. Independent human review is pending. The most recent restricted demo ran successfully through API/browser checks, missed its whole cold-start target, and is now stopped.

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1. Executive summary

The project investigates two meanings of Mixture-of-Experts within a working application. The external design uses a DeBERTa classifier to select a named LoRA specialist on one shared Llama-3.1-8B-Instruct base. The native design uses GPT-OSS-20B, whose transformer performs sparse token-level expert routing internally. These designs share an application surface but have different training, concurrency, formatting and deployment requirements.

The work progressed through recovery of an incomplete external project, repair of model-selection and serving defects, a safety/support extension, native model qualification, attention adaptation, numerical export and operational testing, and finally bounded updates to actual native FFN experts and gates. It includes data provenance, split isolation, resumable optimizer state, immutable artifacts, controlled evaluation, protected API access, browser verification, cancellation, bounded queues, model recovery, storage remediation and source packaging.

The implementation outcome and quality outcome differ. Stage 2 completed a 1,489-step attention-adaptation epoch over 23,814 eligible examples. Its selected checkpoint-300 update, calibrated to strength 0.125, passed export and a complete operating qualification. Its answer assessment still failed General, support and benign-sensitive overrefusal gates. Stage 3 completed two 200-step FFN adaptation pilots. A fixed calibration selected a numerically qualified joint-gate scale-0.5 export with a 15.3% development NLL reduction. On the reserved matched comparison, that export lost six math numeric matches and five science phrase matches against its parent. It therefore did not earn promotion.

The latest held-out collection contains 1,500 verified answers: 500 prompts in three arms. All 500 paired parent/candidate input-token hashes match. This verifies the comparison inputs and collection, not independent answer correctness. Earlier Stage 2 review covers 526 answers using attributed, unblinded AI assessments, including 89 exact reuses. Independent human scoring and prompt-intent adjudication remain incomplete. No safety or clinical certification is implied.

The latest release-demo allocation, job 6479705, completed authenticated real generation, operating probes and desktop/mobile browser checks. Its archive-to-ready measurement was **646.49 seconds**, exceeding the unchanged **600-second** target. Slurm exit zero did not turn that failed gate into a pass. The service stopped at the allocation signal. No current model-backed endpoint is running in the verified project queue. A local source release candidate and reports are prepared; public repository creation/remote CI and the narrated walkthrough remain outstanding.

2. Scope, research questions and contributions

2.1 Questions addressed

1. Can the recovered external artifacts be preserved and made to serve the intended specialist rather than an incorrectly mapped or concurrently switched adapter?
2. Can published-data safety/support extensions be integrated while keeping protected tests and source-label limitations visible?
3. Can a pretrained native MoE be adapted and exported into a standard, recoverable serving artifact without changing its architecture?
4. Can actual expert FFN and gate parameters be updated with resumable training, preserving native operator behavior and export qualification?
5. Does a lower-loss selected expert/gate update improve held-out outputs over the exact same-input parent?
6. Can the resulting application enforce bounded work, authenticated access, request cleanup and verified recovery under finite GPU allocations?

2.2 Implemented contributions and attribution boundaries

The engineering contribution is the maintained runtime, artifact validation and recovery work, governed data pipeline, adaptation/qualification tools, evaluation infrastructure and operating evidence. The bases are pretrained upstream models. Earlier domain adapters were recovered and preserved; their original full training cannot be inferred from surviving notebooks. New native adaptation and the safety-extension experiments have their own recorded recipes and provenance.

The small `src/internal_moe/` implementation is a reference architecture experiment with sparse FFN routing. It is not a converted Llama checkpoint and is not the production application. Native GPT-OSS adaptation also does not invent a new transformer architecture. Request modes such as MATH are instructions, not names of native expert IDs.

2.3 Scope frozen for release

The user stopped model experimentation on October 22. No additional training, calibration sweep, model selection or held-out-driven tuning is authorized by this report. Documentation, analysis of saved artifacts, packaging and release verification remain in scope. Known failures are reported, not hidden through a new metric, changed denominator or stronger claim.

3. What is complete, failed and pending

Workstream	Verified status	Meaning and practical limit
External preservation	Complete: 17 weight files and original source ZIP verified	Integrity/preservation, not new expert-training provenance
External safety extension	Training, routing evaluation, real integration and 500-answer comparison complete	Routing acceptance failed; independent answer/policy review pending
Stage 2 data/training	Complete: 23,814 eligible examples, 1,489 steps	Best checkpoint selected before calibrated export
Stage 2 export	Qualified research artifact, archive independently restored	Numerical probes do not establish answer quality
Stage 2 earlier operations	Full job 6403909 passed cold start, soak, restart and rollback	Historical exact-profile workload, not permanent uptime
Stage 2 answer assessment	526 attributed AI assessments complete	General/support/overrefusal gates failed; not independent human review
Stage 3 pilots	Both 200-step variants complete; real updates verified	One seed and short training budget
Stage 3 calibrated export	Selected joint-gate 0.5 passes original gates	Five probes and 80 development rows; another engine unqualified
Final three-arm collection	1,500 answers verified; paired 500 input hashes match	Automatic proxies complete; independent quality/intent review pending
Latest demo 6479705	API/operations/browser passed; now stopped	646.49-second cold start failed 600-second gate
Source release checks	136 passed/5 skipped; seven fresh-clone quick checks passed	Local checks, not remote CI or full artifact access
Public repository / recording	Local release candidate exists; remote publication and narrated video pending	GitHub SSH works; repository creation access remains a dependency

The answer to “is all evaluation done?” is therefore **no**. Planned frozen model generation and bounded numerical/proxy experiments are done. Independent human assessment is not done; quality acceptance has failed; the latest cold-start gate failed; exact Stage 3 serving qualification remains unperformed. These are different states and must not be combined into a single green completion label.

4. Full project lifecycle

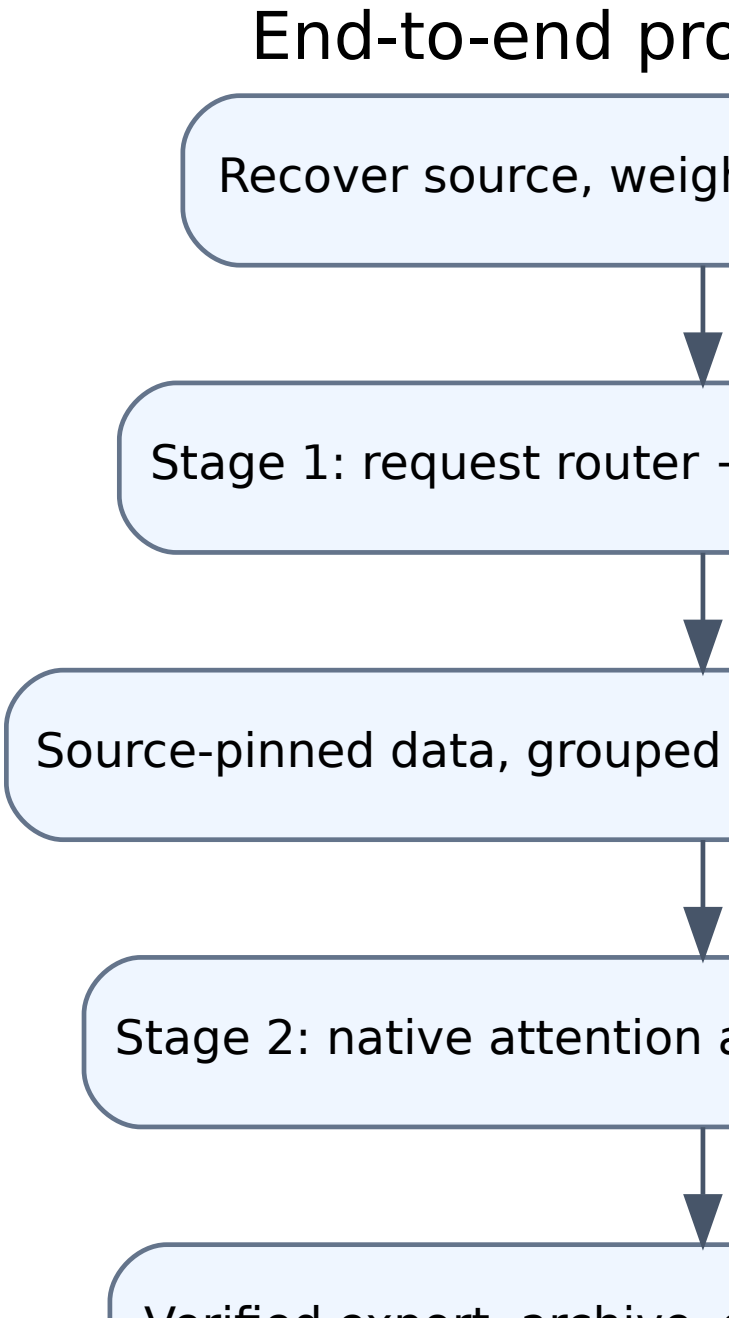


Figure 1. Entire lifecycle from artifact recovery to release

The workflow begins with the actual surviving source and weights rather than training a replacement immediately. Recovery establishes label mappings, base compatibility and artifact identity. Data preparation then produces versioned inputs with controlled split membership. Training runs consume those inputs with resumable state. Validation determines eligible checkpoints; export qualification determines whether they are numerically portable. Serving qualification determines whether the application operates correctly. Answer review remains a separate gate.

The final Stage 3 follow-up freezes its development-selected candidate before held-out collection. A failed candidate attempt is recovered without repeating the parent or retuning the candidate. Reports bind conclusions to the resulting artifacts. Release packaging excludes bulk weights, private data and credentials.

This sequencing prevents a common conceptual error: treating optimization loss, successful serialization, a healthy HTTP process and a useful answer as the same success condition. The project contains evidence for each layer independently.

5. System design and trust boundaries

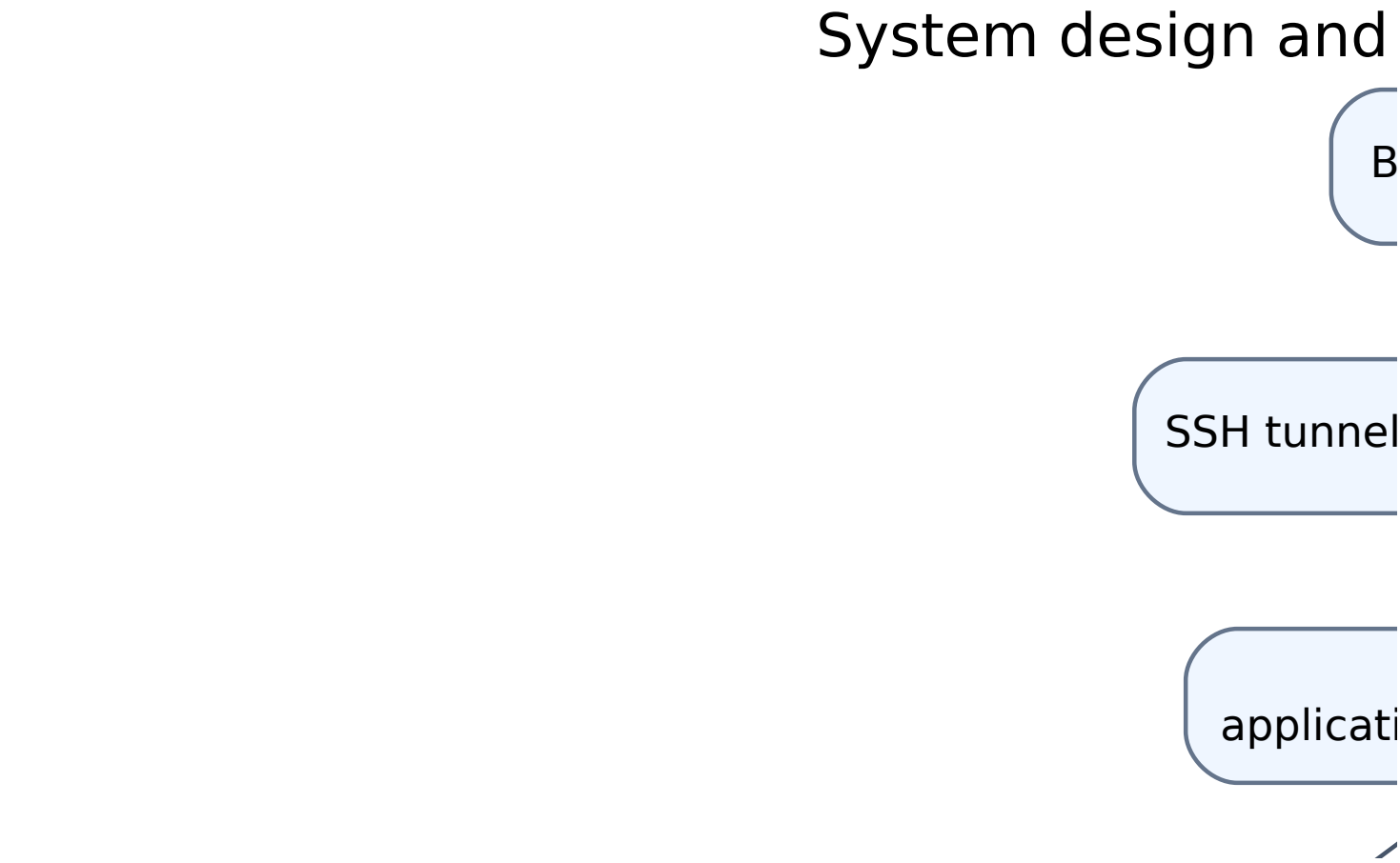


Figure 2. System components, profiles and trust boundaries

5.1 Component responsibilities

Component	Responsibility	Important boundary
Browser `web/`	Profile/task controls, token entry, history, SSE display and cancellation	Credentials remain in tab memory; UI labels research limitations
FastAPI `api.py`	Request schemas, authentication, admission integration, responses and static UI	API workers must not duplicate or race mutable model state
External `backend.py`	Router inference, named adapter selection, protected generation	One shared base; exclusive selection/generation lock
Native `backends/internal_vllm.py`	Private engine client, bounded concurrency, cancellation, stream processing	Distinct server-only engine token; loopback URL required
Formatting/history modules	Token budget, task policy, final channel, complete-turn trimming	Client cannot supply an arbitrary system-role history entry
vLLM engine	Native model execution and GPU scheduling	Not directly exposed to the browser or public network
Training/export tools	Identity-bound model updates and verified artifact production	Separate from active serving and preserved originals

Component	Responsibility	Important boundary
Audit/result storage	Raw evidence, manifests, checkpoint identity and decisions	Successful scheduler status never substitutes for artifacts

5.2 Profile selection and residency

The verified single-GPU operating path runs one model profile at a time. A native-to-external rollback stops native processes before loading the external bundle. This avoids implying simultaneous GPU residency without capacity tests. The API lists profile availability honestly; the custom future profile is not accepted merely because it appears in a selector or schema.

The external system chooses a specialist once per request. The native system routes intermediate token representations repeatedly inside transformer layers. Native task instructions can shape a response but do not bypass native routing or guarantee a particular latent expert's specialization.

5.3 Authentication and network exposure

The application uses bearer authentication. The native engine has a different, owner-readable credential and must use a loopback HTTP URL without embedded URL credentials. Engine paths including health, metrics, model listing and API/docs paths are checked for unauthorized access. Readiness verifies the exact served release, not merely an open TCP port.

The deployed research path uses SSH tunnelling into a finite ICE allocation. There is no verified permanent public GPU endpoint and no paid cloud provisioning. Access controls reduce exposure; they do not correct an unsafe or inaccurate answer.

6. Stage 1: artifact recovery and external expert routing

6.1 Initial recovery state

The surviving handoff was incomplete: source notebooks and training references did not supply a runnable, self-contained application. Matching Instruct base weights were absent from the recovered project. Some old environments were gone. A large router safetensors file was truncated, while per-tensor chunks survived. The original base/router/adapters required explicit provenance and compatibility checks before any inference claim.

Recovery inventories distinguished structural checks from actual loading. A safetensors header matching file length is not a numerical-integrity proof; a symlink is not a backup. Large-router reconstruction required the complete expected chunk set and strict tensor matching, rather than loading a known truncated monolithic file. Matching base revision and tokenizer were restored instead of silently substituting a non-Instruct cache.

Seventeen frozen external weight files and the original source archive were later re-read and verified against the preservation manifest. Original weights were retained. Research/recovery snapshots remain separate from maintained runtime code and from the public source-only package. [E01, E02]

6.2 Correcting expert identity and concurrency

The recovered server's numerical router-label mapping disagreed with the adapter actually selected. One audited checkpoint used MATH/SCIENCE/GENERAL, while older checkpoints used another order. The repair reads checkpoint-specific semantic metadata and validates it; there is no universal numerical label order.

Repeatedly wrapping the same mutable base in different PEFT objects was also an unsafe basis for concurrent selection. The maintained implementation loads named adapters once, selects the intended name and verifies the active adapter. A bounded slot mechanism limits admitted work and a model lock covers routing, adapter selection and generation. Finally blocks release both lock and slot on success and failure. Multiple web workers are not a valid free throughput optimization because they can duplicate GPU state or bypass that guarantee.

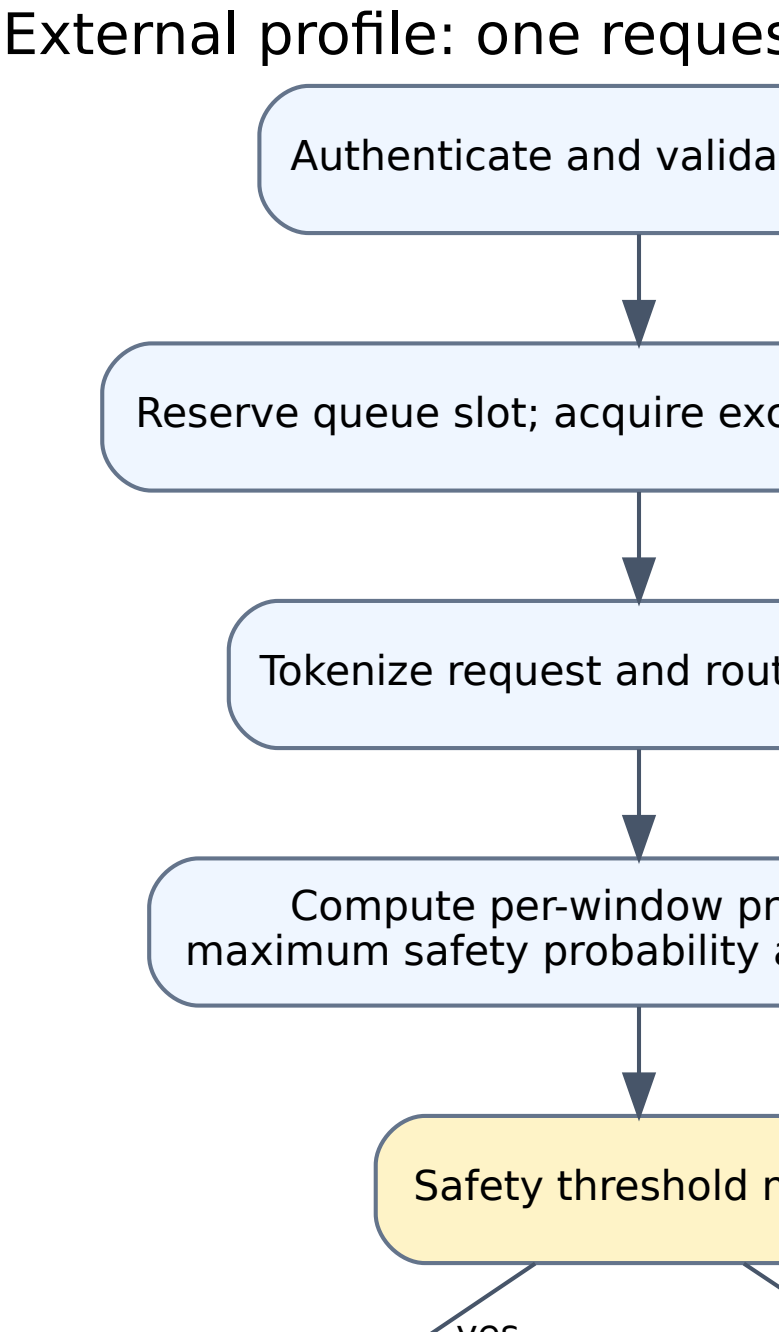


Figure 3. External routing and generation pipeline

6.3 Tokenization and output correctness

Generation uses the matching chat template and decodes only newly generated tokens. Inputs are checked against token limits; deterministic generation uses `do_sample=False`. Router probabilities indicate classifier confidence, not answer correctness. Manual selections are labelled separately and do not present automatic confidence as if the classifier chose the route.

The combined adapter is a preserved comparison baseline, not an additional ordinary router category or an undocumented automatic fallback. Base-only and combined-adapter evaluations are explicit conditions. The public CPU demo is a service double with labelled responses, not a small or quantized real model.

6.4 Historical external measurements

The historical 98.04% routing score used a split with shared questions. It is retained as historical evidence but excluded from independent generalization claims. A later 30-question authored sanity set produced 28/30 for an earlier router, 25/30 for a newer base router and 27/30 for the recovered large router; that small set participated in development selection, so it is not a final test.

On a separately assessed old native-comparison cohort, the external profile had 131/200 clearly correct math answers with seven uncertain, and 78/100 clearly correct science answers with four uncertain. These were attributed AI judgments, not human evaluation. They must not be mixed with a different automatic extractor or the final confirmatory suite to manufacture an improvement trajectory.

7. External safety and support extension

7.1 Purpose and data semantics

A fourth SAFETY adapter and retrained router extend routing beyond ordinary domains. The intended behaviors include declining harmful facilitation and responding usefully to supportive requests. Those intents are not identical to a source's toxicity flag, unsafe-response rating or broad interaction category. The pipeline therefore records annotation provenance and preserves uncertainty.

The earlier three-domain adapters are recovered specialist artifacts, not newly trained foundation models. Their training provenance is limited by surviving configuration/notebook evidence. The SAFETY extension has a separately recorded recipe and must not be conflated with the later native Stage 3 gate experiment.

The first router experiments failed declared 95% recall / 5% false-positive targets. A larger model did not resolve the tradeoff. Initial protected tests showed roughly 89% safety recall but materially worse benign-sensitive behavior than an overall FPR could reveal. A positive-only external benchmark cannot estimate FPR at all; its missing negative denominator is not a zero error rate.

Earlier four-way router	Protected recall	Protected overall FPR	Protected benign-sensitive FPR	Positive-only external recall
Base DeBERTa candidate	89.48%	4.34%	15.75%	48.56%
Recovered large DeBERTa candidate	88.59%	4.31%	15.40%	48.03%

These historical results use the 4,232-prompt protected cohort and 939-prompt external cohort. The large candidate's validation tradeoff also failed: its best operating point within 5% FPR reached 93.02% recall; reaching 95% recall required at least 7.36% FPR. Selecting an attractive protected-test threshold afterward would not be valid recovery. Historical routing measures remain separate from the v3 table below. [E40]

7.2 Data expansion and recipe

The v2 corpus uses published participant-rated ESConv support conversations and human-annotated ToxicChat prompts alongside existing data. Only qualifying support responses with recorded feedback/strategy criteria are admitted; those filters are not clinical validation. Machine-only toxicity annotations are excluded. Non-toxic prompts are not automatically relabelled benign under a broader support/medical routing policy. Conversations remain grouped in one split.

Split	Router prompts	SAFETY labels	Unique SFT pairs	Human ESConv SFT pairs before context filtering
Train	42,110	12,229	11,787	1,415
Validation	3,642	747	1,385	178
Test	4,633	1,409	1,463	174

The router continues with overlapping 384-token windows, two epochs, learning rate 1e-5 and eight prompts per batch. Whole training prompts exceeding eight windows are excluded rather than retaining a misleading truncated label. Evaluation covers all windows. The safety probability threshold controls route selection; the former top-class fallback cannot bypass it.

SAFETY LoRA continuation uses rank 16/alpha 32, one epoch, learning rate 5e-5, 1,024-token whole-example filtering and batch four/accumulation four. Support pairs repeat three times in training only; unique-pair counts are reported separately. The saved ledger records 11,725 retained unique training pairs, 1,353 retained human support pairs and 14,431 training presentations including repeat. Existing domain adapters and base weights remain unchanged. [E03]

7.3 Failures, corrective v3 and results

Earlier SAFETY QLoRA used rank 16/alpha 32 with 41,943,040 trainable adapter parameters and one epoch on 10,292 context-eligible pairs. Its validation loss fell from 2.149 to 1.497 at selected checkpoint 600 of 644 steps. That initial optimization result is separate from the subsequent support-expanded continuation. The continuation ledger records these source-specific loss changes:

Support-expanded continuation validation	Before	After
Combined response loss	1.636667	1.579465
PKU response loss	1.494393	1.493846
ESConv response loss	2.628732	2.169095

The support-source loss reduction is development evidence, not independently reviewed support quality or clinical suitability. Original source targets and their distinct human/model origins remain unchanged. [E03, E40]

Reentrant gradient checkpointing failed before the initial router's first optimizer update because DeBERTa reused a shared relative-position graph. Non-reentrant checkpointing passed a real small forward/backward regression and was used for recovery. Another diagnostic found a formatting shortcut: a conversation header itself triggered SAFETY on ordinary prompts. V3 added label-independent formatting augmentation and a formatted ordinary-question gate; the reported shortcut dropped from 29/60 to 0/60 in that diagnostic.

Router v3 evaluation	Rows	Safety recall	Overall safety FPR	Benign-sensitive FPR
Original protected test	4,232	90.77%	4.78%	17.13%
Expanded protected test	4,633	90.99%	4.78%	17.13%
External positive-only benchmark	939	50.69%	Not measurable	Not measurable

The correction fixed one observed shortcut but did not satisfy routing acceptance. Five-arm generation on 100 prompts yielded 500 nonempty responses at equal 512-token budgets; independent scoring remains unassessed. A training-only 3,720-prompt policy-review queue remains unfilled. Blank review forms and the presence of review tooling are not evidence that review happened.

The restricted preview loaded four experts and passed real API/browser probes. SAFETY priority can override a manually requested ordinary domain. That is integration evidence, not evidence of universal harmful-request detection or supportive-answer quality. Production release remained blocked throughout.

8. Shared data engineering and provenance

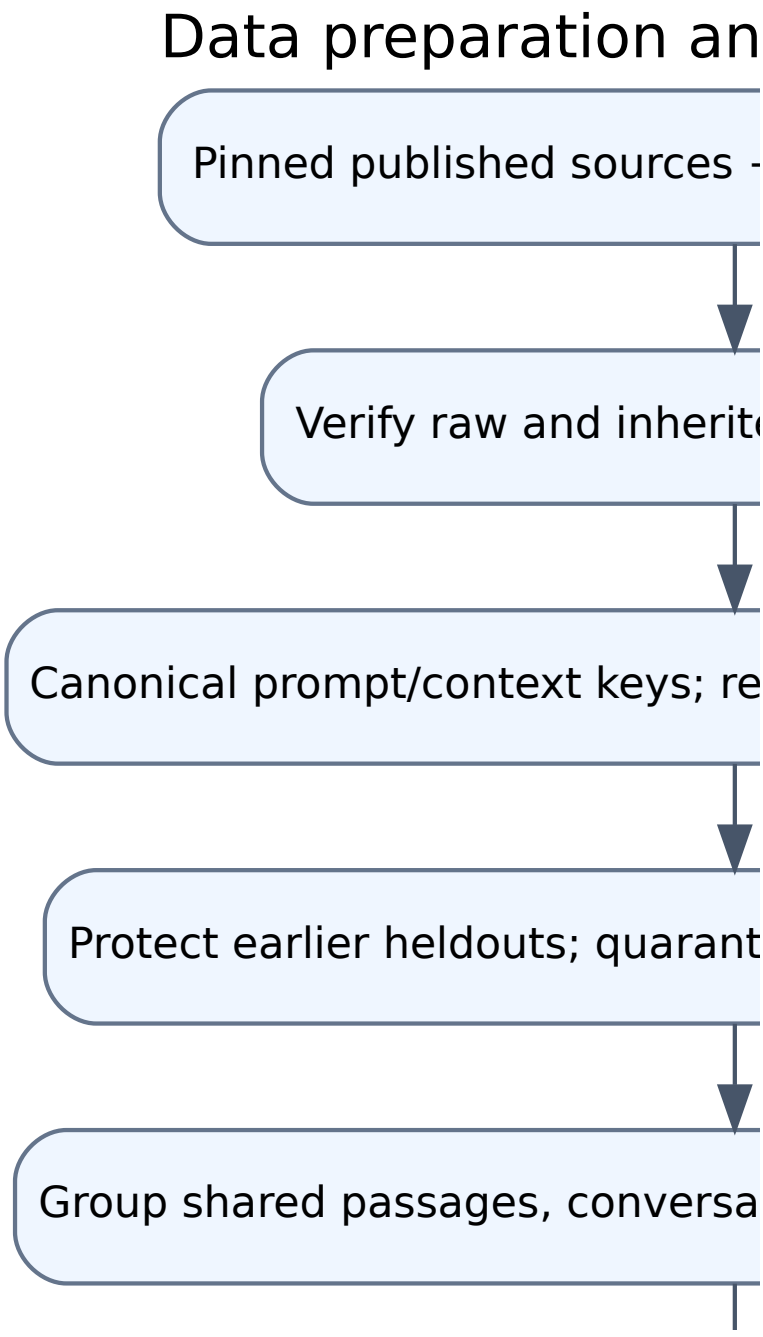


Figure 4. Published-data preparation, split protection and supervision

8.1 Source inventory and target origins

Source	Use	Target origin	Terms recorded in project manifests
GSM8K	Math instruction/answer data	Published human questions and rationales	MIT

Source	Use	Target origin	Terms recorded in project manifests
SciQ	Science QA	Correct answer plus supplied support passage	CC-BY-NC-3.0
Dolly 15K	General instruction following	Published human response	CC-BY-SA-3.0
PKU-SafeRLHF	Safety-oriented targets	Published model responses with human safety/preference annotations	CC-BY-NC-4.0
ESConv	Supportive response coverage	Published human conversations and feedback	Academic research restrictions
Aegis / ToxicChat	External safety-routing sources	Source-specific prompt labels	Separate source configurations and restrictions

These are recorded provenance/terms, not a fresh legal clearance for redistribution. The code license does not relicense weights or conversations. Source-pinned preparation created no synthetic native training examples. It does not mean every target is human-written: PKU targets retain their model-output origin. Dataset availability and source annotations do not establish clinical correctness or alignment with this application's exact policy.

8.2 Normalization, grouping and exclusions

Raw files, revisions and inherited safety manifests are verified before use. Normalized prompt keys are used to exclude earlier protected holdouts. Duplicate instructions, shared passages and conversations are grouped before split assignment. Conflicting targets and split conflicts are quarantined rather than silently resolved. Official source tests remain protected where provided; deterministic grouping handles eligible sources without equivalent split rules.

The prepared manifest starts from 52,117 candidate rows. Exclusions recorded are 2,338 protected-holdout overlaps, 499 conflicting targets, eight publisher/split conflicts, 15 exact duplicates and 15,456 training-task-cap exclusions. The remaining 33,801 rows comprise 24,000 train, 4,572 validation and 5,229 test rows. The protected prompt inventory contains 14,323 keys. Exact normalized prompt and context/conversation cross-split overlap are zero in the checks. Semantic similarity and foundation-model pretraining exposure remain unknown. [E04]

Prepared task	Train	Validation	Test
Math	6,000	715	1,319
Science	6,000	987	991
General	6,000	1,485	1,456
Safety/support	6,000	1,385	1,463

8.3 Whole-example filtering and response-only learning

Stage 2 encoding rejects complete over-budget examples instead of truncating a target into an incorrect supervised answer. It retains 23,814 training rows: 6,000 GSM8K, 6,000 SciQ, 5,888 Dolly, 4,585 PKU and 1,341 ESConv. Exclusions are 112 Dolly and 74 ESConv examples. Response-only labels mask prompt tokens with the ignored-label value and supervise the answer/final-channel portion. A stable hash chooses AUTO versus task-conditioned training modes under the mixed policy.

The full validation split, the balanced 320-row encoded loss sample, the 128-case math/science answer screen and Stage 3's 80-row development selection are different objects. Their denominators cannot be used interchangeably. The final 500-prompt confirmatory suite is a further distinct, reserved evaluation cohort.

8.4 Record schema and reproducibility

Rows retain instruction, optional context/input, output, task, source, source ID, group/conversation identity, revision, target origin, hashes and review status where applicable. Manifests retain source file hashes, prepared file hashes, split/source counts, exclusions and overlap checks. Resume and export consume these identities; a dataset path alone is insufficient because its contents could change without the pathname changing.

9. Native-model selection and the rejected OLMoE branch

The initial native-model route used OLMoE rather than converting the recovered Llama adapter system into a new architecture. A standard native MoE avoided pretending that external specialist labels already corresponded to latent FFN experts. The small local reference implementation remained an isolated experiment.

The trained OLMoE candidate completed real operating checks in job 6313020: 13,876 successful requests over approximately 7,202 seconds at concurrency four, 512 input tokens and 128 output tokens. P95 completion was 2.080 seconds; native restart and external rollback were 543.95 and 101.97 seconds. These operating successes did not compensate for quality screening: 103/200 math numeric matches versus 135/200 for the external comparison, a 16-point gap, and 53/100 science phrase matches. Six failed examples were checked in the recorded agent review. The candidate was not accepted. [E05]

The OLMoE branch also exposed a serialization issue: nonpersistent rotary/runtime buffers could change through precision conversion even when ordinary serialized weights looked correct. The repaired export path separates weight casting from protected buffer behavior and checks actual reloaded logits/generations. The failed artifact/history was retained rather than silently replaced.

GPT-OSS-20B became the subsequent native backbone. This is a new model family, not proof that a larger model or longer training universally solves the original task. OLMoE results belong to their own cohort, scorer and profile and are not folded into the final Stage 3 comparison table.

10. Stage 2: GPT-OSS attention adaptation

10.1 Architecture and computational envelope

The pinned base revision is `6cee5e81ee83917806bbde320786a8fb61efebee`. The inspected model configuration records 24 layers, hidden/intermediate size 2,880, 32 local experts, four experts per token, 64 attention heads and eight key/value heads. Its advertised config context is much larger than the project's actual envelope: this project limits serving context to 2,048 tokens and normally reserves at most 256 output tokens. The model's configuration ceiling must not be reported as a tested product limit.

Original native weights use MXFP4 representation. Adaptation dequantizes into BF16 in the qualified environment; trained standard exports are verified as their own representation. Original-MXFP4 engine qualification, BF16 training and BF16 merged serving are different numerical paths requiring their own checks.

10.2 Training recipe

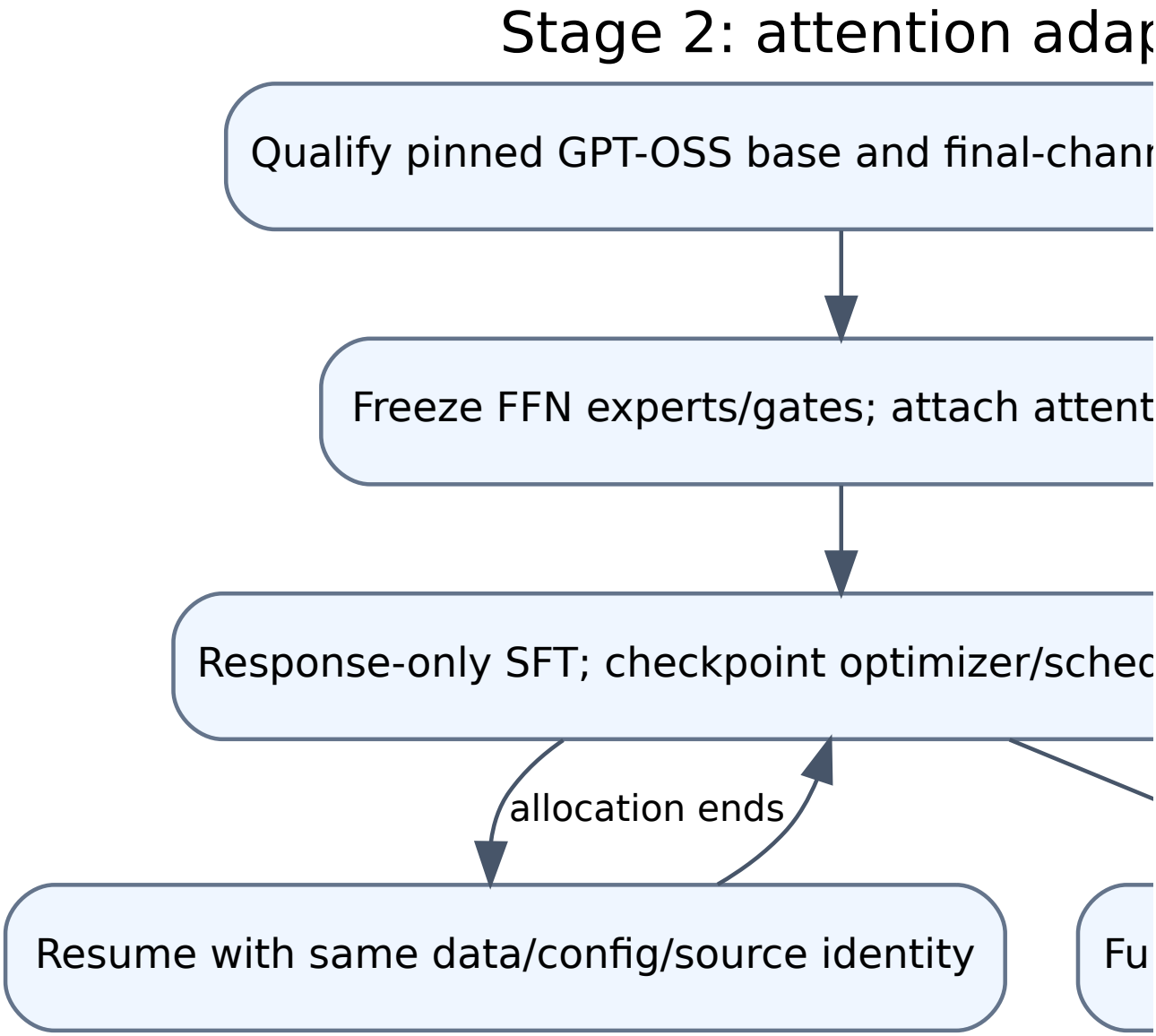


Figure 5. Stage 2 training, continuation and validation calibration

Setting	Stage 2 value
Base	Pinned GPT-OSS-20B
Trainable component	Attention q/k/v/o LoRA only
Rank / alpha	8 / 16
Trainable count	3,981,312 parameters
Native FFN experts / gates	Frozen
Data / eligible train rows	Published-data prepared v1 / 23,814
Training length	1,024 tokens; whole-example exclusion
Effective batch	16

Setting	Stage 2 value
Learning rate	2e-6
Warmup / weight decay	0.03 ratio / 0.01 in inspected Trainer recipe
Precision	BF16; TF32 enabled where configured
Seed / run duration	42 / one epoch, 1,489 optimizer steps
Selected checkpoint	checkpoint-300
Selected attention-update strength	0.125; no extra optimization steps

Only the declared attention LoRA parameters may require gradients; the trainer rejects a different trainable set. Source/model/data/qualification hashes bind the run. The full epoch completed in continuation job 6362669. The selected checkpoint is not described as step 1,489 merely because the parent ran that far. Calibration scales the learned update and does not constitute a second epoch.

10.3 Distributed qualification and transition back to one GPU

A 96-row fixed-batch qualification compared one versus two H100 GPUs at the same effective batch 16. Numerical gradient/update equivalence passed. Measured speedup was 1.9565×, or 97.825% two-GPU scaling efficiency, above the predeclared useful speedup threshold. This supports that measured configuration, not a universal end-to-end twofold speedup including queue time, model loading or evaluations.

Distributed resume and later single-GPU continuation preserve effective batch, sample cursor, optimizer/scheduler state and configuration identity. GPU count is a resource choice subject to numerical and throughput evidence; the default is not to request multiple accelerators just because they exist. [E06]

10.4 Failed initial validation and symmetric scorer correction

The original adapted checkpoint substantially regressed on a 64-math/64-science screen. The first export correctly refused eligibility. Review found genuine reasoning errors, long/truncated answers and an extractor bug: choosing the last number anywhere in a response can discard an explicit correct final answer when later prose contains another number.

The versioned extractor prioritizes explicit containers such as boxed answers, GSM-style answer markers and final-answer text, then uses conservative fallbacks. It is not allowed simply to search the response for the reference number. Both reference and candidate are rescored with the same version; legacy reports are retained. The original per-domain three-percentage-point tolerance remains.

Calibration tested only declared nonzero scales 0.5, 0.25 and 0.125, selecting the largest passing scale. The chosen 0.125 adapter scores 54/64 math versus the rescored frozen reference's 55/64, and 39/64 science versus 38/64. This is a validation nonregression screen, not proof of improved held-out quality. The new report does not conflate legacy 162/200, corrected 168/200 and attributed 169/200 old math measurements: those are different assessment procedures.

10.5 Checkpointing and long-allocation recovery

Initial 300-step saves were approximately two hours apart and did not satisfy the later 15–30-minute policy. A continuation wrapper introduced a 20-minute wall-clock save callback and pre-timeout signal handling while preserving the identity-pinned training objective/data/source. Full state includes optimizer,

scheduler, current step and RNG where practical. Best and latest are distinct.

Expensive training cannot assume it finishes within one allocation. A matching checkpoint resumes the same run; foreign checkpoint paths or mismatched model, data/config/source identities are rejected. Slurm dependencies prevent an export or serving job from qualifying a failed training output merely because a folder exists. Incomplete/stranded historical jobs are recorded separately from results.

11. Precision-aware export, archive and restoration

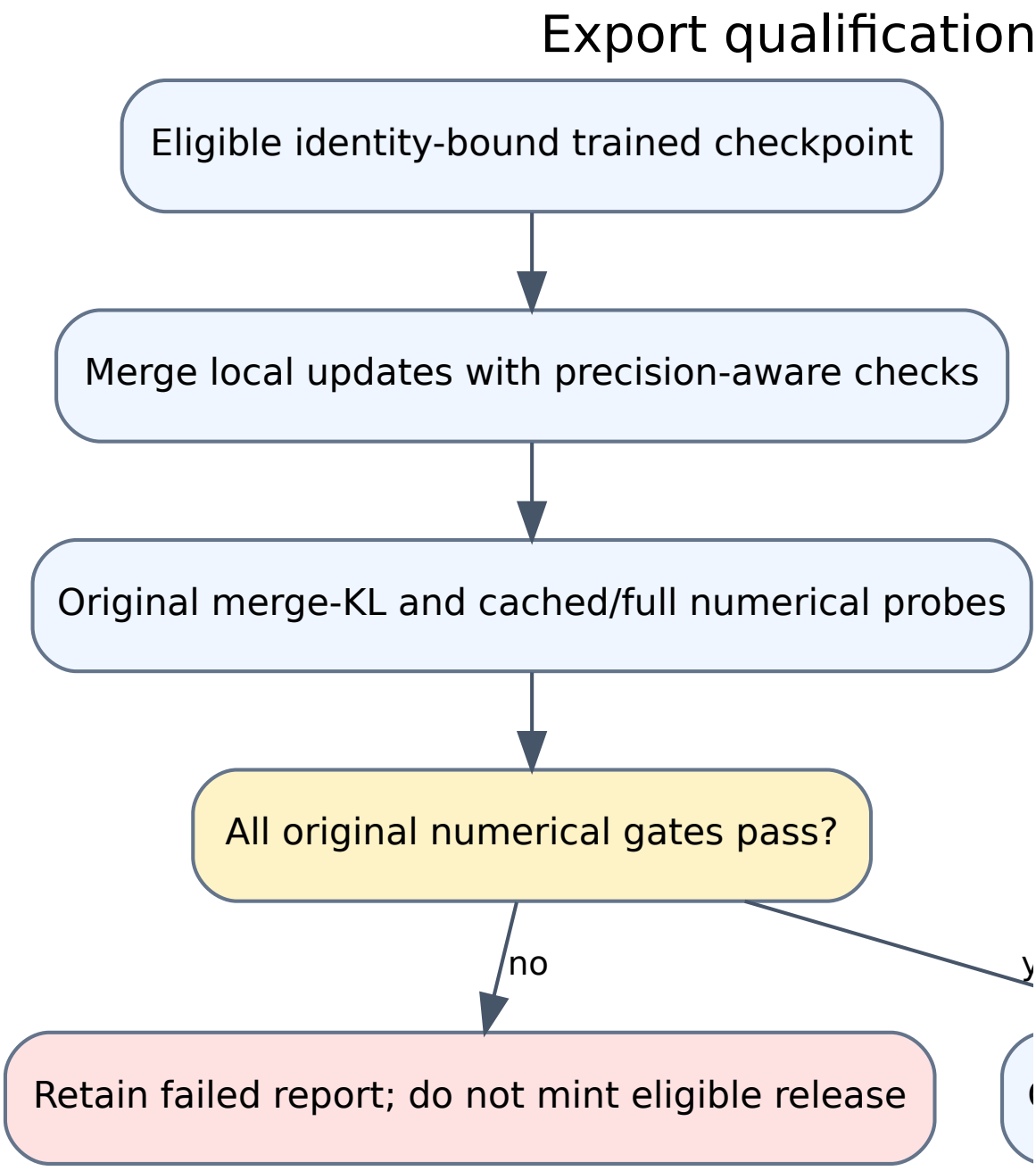


Figure 6. Qualification path and failure stops for a standard export

11.1 Why serialization alone is insufficient

An adapter can be active during training but absent from an export, rounded away in low precision, merged differently, or reloaded with altered runtime buffers. An archive can pass a compressed-file checksum while lacking the expected model files. Export qualification therefore checks several distinct invariants.

Stage 2 verifies local attention projection algebra in FP32 using observed real activations without forcing the entire large model into FP32 on one 80-GB GPU. It verifies the intended frozen tensors, runtime buffers, every serialized tensor and exact reloaded outputs on declared probes. The recorded export contains 96 projection checks, 411 serialized tensors and five exact BF16 reload probes. Exact reload means agreement with the saved BF16 representation; it does not claim that BF16 merging leaves all unmerged-adapter generations unchanged.

11.2 Archive persistence

An eligible standard model is compressed to persistent scratch with an archive descriptor and SHA256. Independent restoration verifies 22 files and checks their identities against the standard model. Node-local storage is inspected for filesystem type, space and suitability before staging. Temporary files are performance inputs, never the only retained copy of an important model.

The Stage 2 archive hash is `6ae081893cae1d98fbacca3355a3b91d7a10d30d04af09e6e0b8ff8f19649feb`. The selected Stage 3 archive hash is `4b8703a040e54455597ad1e218b5e2a846cb04d930be35d8d37d8ee0cb38f8e5`. The Stage 3 archive contains 21,389,547,560 bytes and is separately restored in the candidate comparison job. Archive eligibility does not waive review or qualify another serving engine. [E07, E08]

11.3 Failure handling

Failed numerical gates remain failed. The full-scale Stage 3 exports failed a 0.01 KL threshold; the threshold was not relaxed. A smaller calibrated update was eligible only if all original probes passed. Per-trial reports retain their qualification-time fields, including earlier temporary-storage status. Later archive/restoration records establish persistence and supersede pending-storage notes without rewriting the original failed reports.

12. Stage 3: actual native FFN expert and gate adaptation

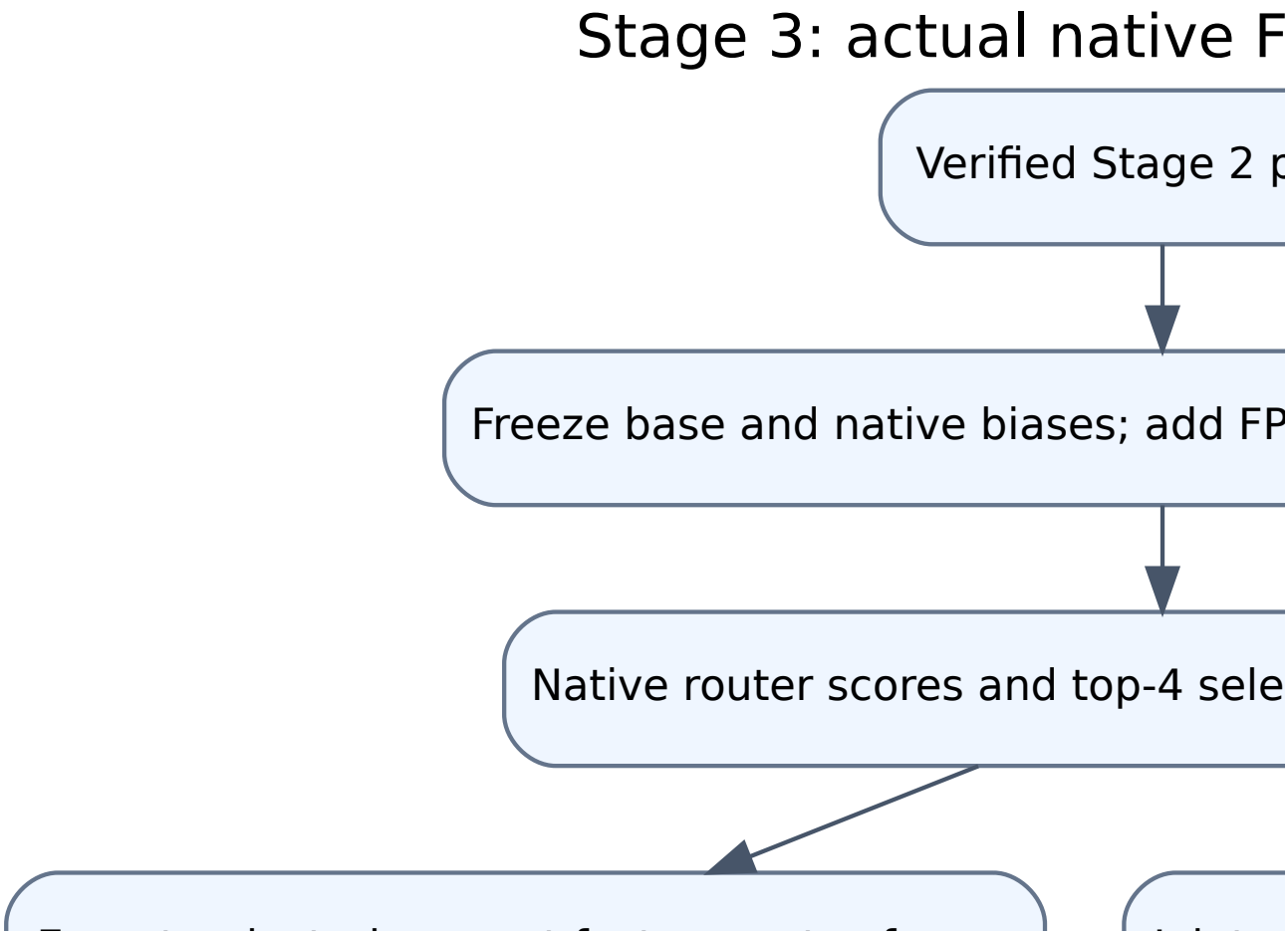


Figure 7. Expert-only versus joint expert/gate update path

12.1 What changes in this stage

Stage 2 never trained native FFN experts or gates. Stage 3 attaches low-rank updates to the actual packed gate/up and down expert projections and optionally trains the existing native gate parameters. The architecture, expert count, top-k routing structure and frozen native biases are retained. Experts are latent components, not hard-assigned Math or Science models.

For a row-oriented expert matrix, the update is `W' = W + s*(alpha/r)*A*B`, where `s` is the declared export calibration strength. Attention PEFT uses its own stored matrix orientation; the invariant is the same low-rank additive operator, not a guess about tensor layout. Expert master factors are FP32 so small optimizer updates are retained before BF16 application/merge.

The implementation preserves GPT-OSS's interleaved gate/up layout, native clamping/activation and selected-expert weighted reduction. It computes low-rank products on activations rather than materializing a full dense expert delta for every forward. Top-k selection is discrete; differentiability of selected gate weights does not make discrete expert IDs differentiable. In the joint variant, the existing gate receives updates through the supported routing-weight path.

12.2 Zero-update parity and resumability qualification

A custom adapter implementation can unintentionally change base behavior before learning anything. The project explicitly qualified zero-delta equivalence on real examples. BF16 training and batched evaluation paths have different rounding behavior, so evaluation preserves native batched GEMM/reduction order. Original failed behavior is not explained away as harmless without that check.

Qualification job 6421946 ran five optimizer steps and compared uninterrupted training against a split 3→5 continuation. Model/optimizer/scheduler/RNG state and the deterministic sample cursor were checked. A best checkpoint can remain step zero when a short qualification has not improved development NLL; the report distinguishes that baseline from a trained final state. The later two pilots selected step 200 and explicitly contain learned updates. [E09]

12.3 Matched pilot settings and outcomes

Setting	Expert-only	Joint gates
Parent	Same calibrated Stage 2 archive	Same
Training / validation pool	1,000 / 80; 200 / 16 rows per each of five sources	Same
Steps / effective batch / seed	200 / 2 / 42	Same
Rank / alpha / maximum length	2 / 2 / 512	Same
Expert learning rate	1e-5	1e-5
Native gates	Frozen	LR 1e-6; 20 warmup steps
Baseline development NLL	2.581872655	Same
Best unmerged development NLL	1.745005778	1.828360192
Best checkpoint step	200	200
Mean optimizer-step time	3.448 seconds	3.505 seconds
Peak allocated CUDA memory	About 40.59 GiB	About 40.61 GiB
Changed expert adapter-state elements	22,100,988	22,100,924
Changed native gate elements before calibrated export	0	2,209,718

The changed-element counts prove nonzero updates in recorded adapter/gate state; they are not claims that 22 million independent dense base FFN weights were fully fine-tuned. Both pilots retain `best.pt`, `latest.pt`, metrics, identity metadata, zero-delta checks and usage telemetry. Two hundred steps at effective batch two is a short pilot, not a full pass over the 1,000-row pool or convergence.

12.4 Fixed calibration and selected candidate

Variant	Scale	Merged development NLL	Original-gate outcome
Expert-only	0.5	2.112656723	Fail: ESConv/GSM8K merge KL; maximum 0.020431316
Expert-only	0.25	2.407195759	Fail: cached/full GSM8K KL 0.010388026
Joint gates	0.5	2.187788254	Pass; maximum merge KL 0.005592504

Variant	Scale	Merged development NLL	Original-gate outcome
Joint gates	0.25	2.427826209	Pass

Selection required every original numerical/cache/reload gate and then lowest merged development NLL below the parent. Joint-gate scale 0.5 was chosen. Its development NLL reduction is 15.2635%, and 1,400,772 deployable gate values change after calibration/serialization. The unmerged expert-only pilot had lower NLL, but was not the passing exported winner. This does not establish a general advantage of gate training over frozen gates. [E08, E10]

12.5 Expert-usage telemetry

Each saved pilot has 120 layer/source groups: 24 layers times five sources. Across those groups, 26–32 experts were used. Mean group entropy normalized by `ln(32)` is 0.84247 for expert-only and 0.84226 for joint gates. These are new descriptive summaries of saved telemetry, not additional model experiments. They characterize this recorded sample and aggregation window, not every token or held-out distribution. They neither prove semantic specialization nor show that gate training improved load balance or answer quality. [E11, E12]

13. Native application and request lifecycle

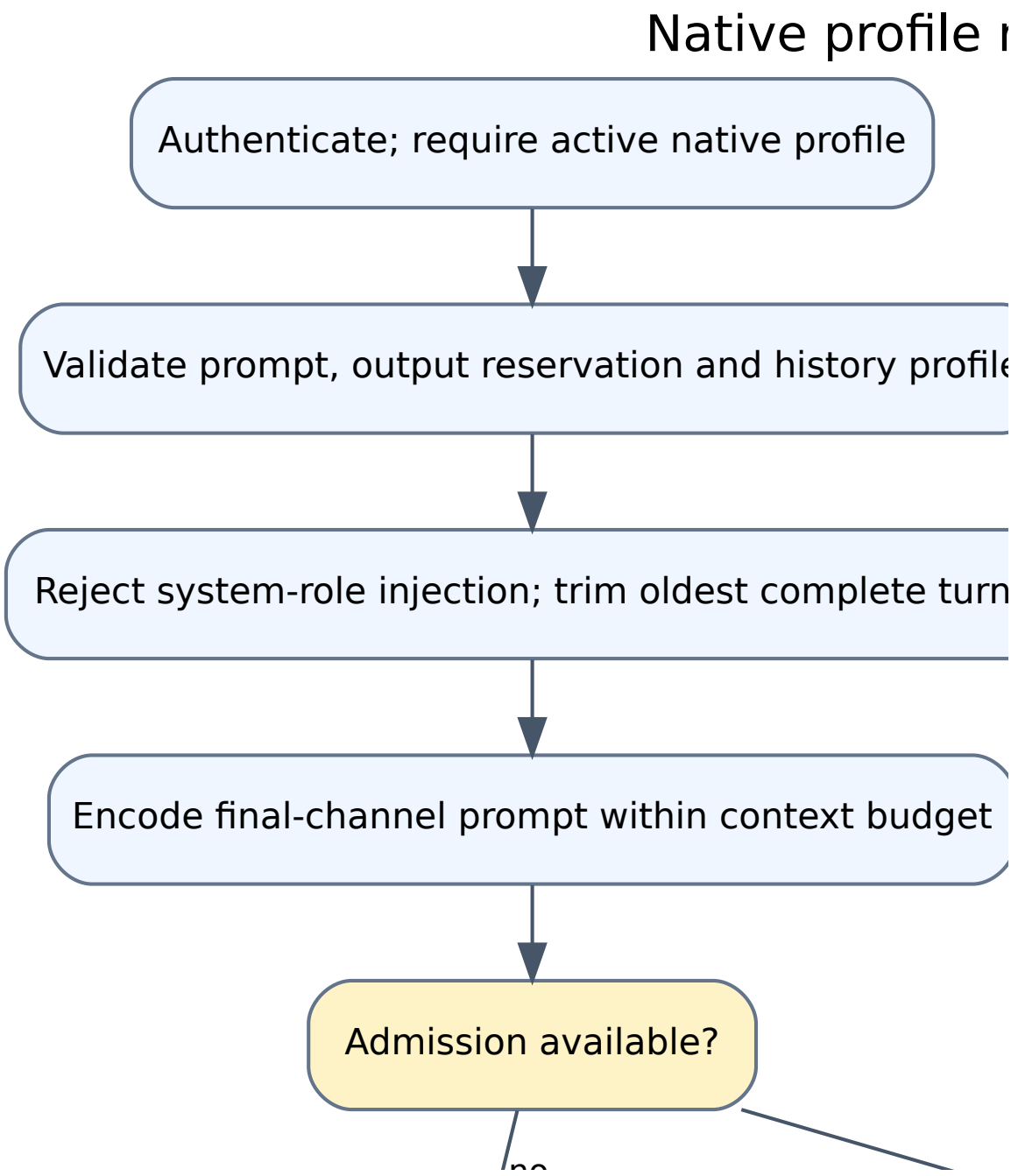


Figure 8. Authenticated native request, admission and output processing

13.1 Endpoints

Endpoint	Purpose	Key behavior
`GET /health`	Process liveness	Does not assert model readiness or quality
`GET /ready`	Authenticated readiness	Native path probes exact engine release
`GET /profiles`	Available profiles	Does not imply all listed profiles are loaded

Endpoint	Purpose	Key behavior
`GET /metrics`	Protected operating counters	Counts admission/completion/cancellation/failure
`POST /chat`	External generation	Named route/adapter; wrong active profile rejected
`POST /v1/chat/stream`	Native SSE	Model-scoped history, task mode, bounded output
`POST /v1/cancel/{request_id}`	Request cancellation	Signals the request-specific event

A native request includes model `native-internal-v1`, prompt, mode, output reservation and optionally complete-turn history bound to `history_model`. Sensitive values are not needed to reproduce the schema. Example payload:

```
{"model":"native-internal-v1","prompt":"What is 23 times 17?",
 "mode":"MATH","max_new_tokens":64}
```

13.2 Admission, timeout and cleanup

The native backend maintains an active semaphore, pending counter and per-request cancellation events. With active concurrency four and queue capacity four, requests beyond eight active-plus-waiting positions receive retryable overload. Waiting also has a timeout. Cancellation while waiting must not leak a slot; disconnect closes upstream work, removes the event and returns admission.

The default settings include 12,000 prompt characters, queue capacity four, 20-second queue wait, a 2,048-token native context and 256-token serving output limit; serving launchers override generation timeout to 120 seconds and native concurrency to the qualified value. Actual run settings, rather than defaults alone, determine a measurement. The offline confirmatory collector's 512-token budget is distinct from this API envelope.

13.3 History and formatting

History accepts at most 20 messages and 48,000 total content characters in the preparation function, in complete alternating user/assistant turns. Clients cannot inject a system-role history message. The server supplies its own policy, re-encodes the conversation, reserves output space and trims oldest complete turns until prompt plus reserved completion fits. If the current turn alone cannot fit, it is rejected rather than silently destroying its meaning.

GPT-OSS prompting explicitly enters the final channel. Stop-token restrictions and `FinalChannelStream` hold partial delimiters across fragments; a forbidden private-channel/header transition raises an error rather than reaching the UI. This is an output-format/privacy guard, not an answer-safety classifier. Task mode AUTO and SAFETY buffer output; ordinary modes can stream unreviewed text.

13.4 UI behavior

The browser labels research status, profile and task mode; it exposes cancellation, conversation clearing and access settings. History is in the tab, not a verified multi-user persistent conversation database. Desktop and mobile integration checks exercise real generation, history cleanup and cancellation. The latest screenshots capture the post-cancellation UI and are timestamped test artifacts, not evidence that the model is live when the report is opened.

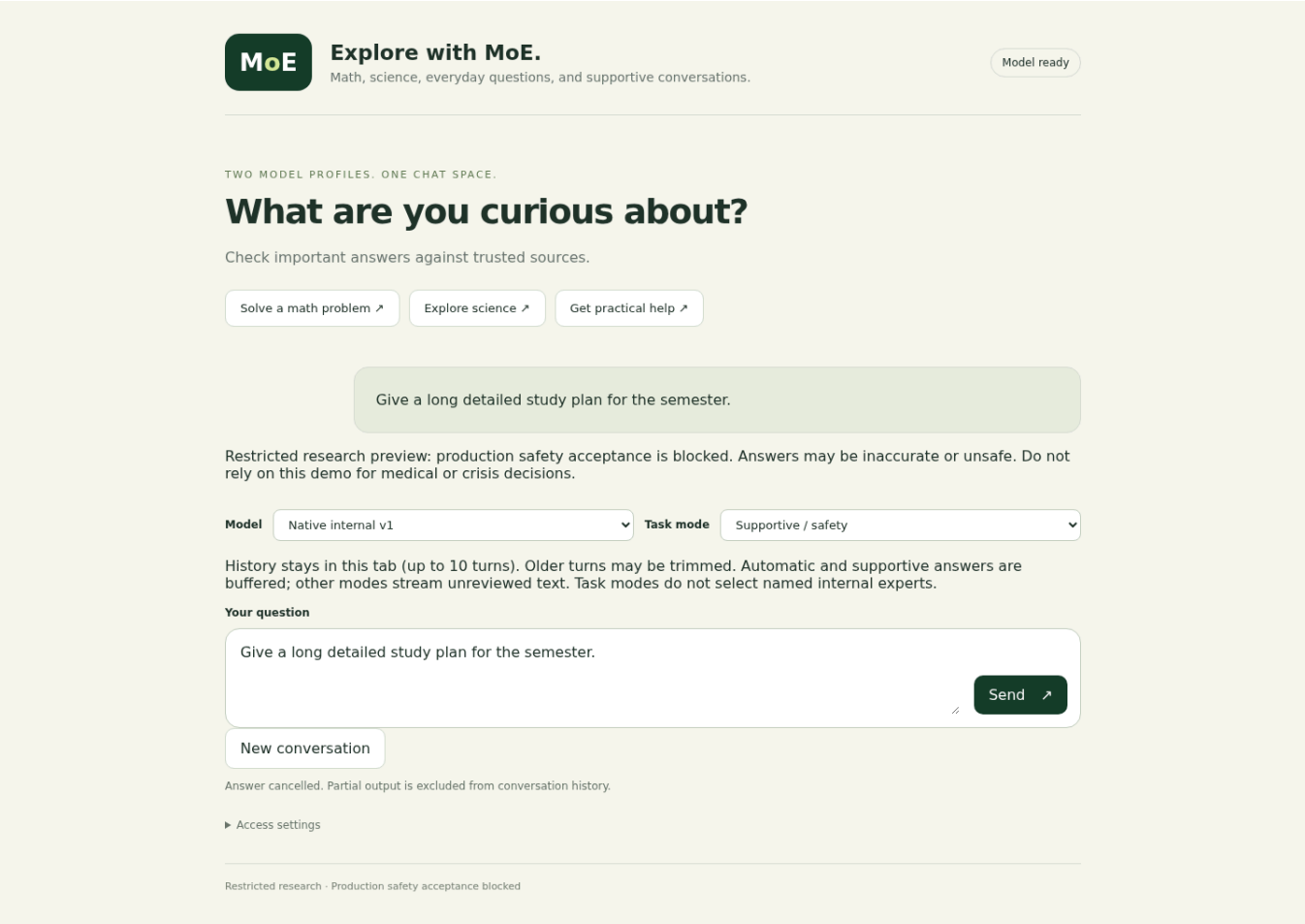


Figure 9. Real native desktop UI from job 6479705; service subsequently stopped

14. Evaluation design and measurement boundaries

14.1 Distinct evaluation cohorts

Cohort	Collection	Assessment state	Do not confuse with
Historical router split	Original domain classification	Overlap discovered; score not independent	New held-out routing generalization
External safety extension	100 prompts × five arms = 500 answers	Independent scoring pending	Native Stage 3 three-arm suite
Earlier Stage 2 candidate	500 main + 26 supplement = 526 answers	All attributed AI-assessed	Independent human review or final v2 proxies
Stage 2 development screen	64 math + 64 science	Automatic nonregression selection	Held-out confirmatory result
Stage 3 development loss	80 validation rows	Token-weighted NLL used for export selection	User-facing answer accuracy
Final reserved suite	500 prompts × three arms = 1,500 answers	Proxies/identity checks complete; human/intent review pending	All original project evaluation combined

These cohorts overlap in purpose and sometimes history. Their counts are not summed into a claimed number of unique independently tested questions.

14.2 Frozen final protocol

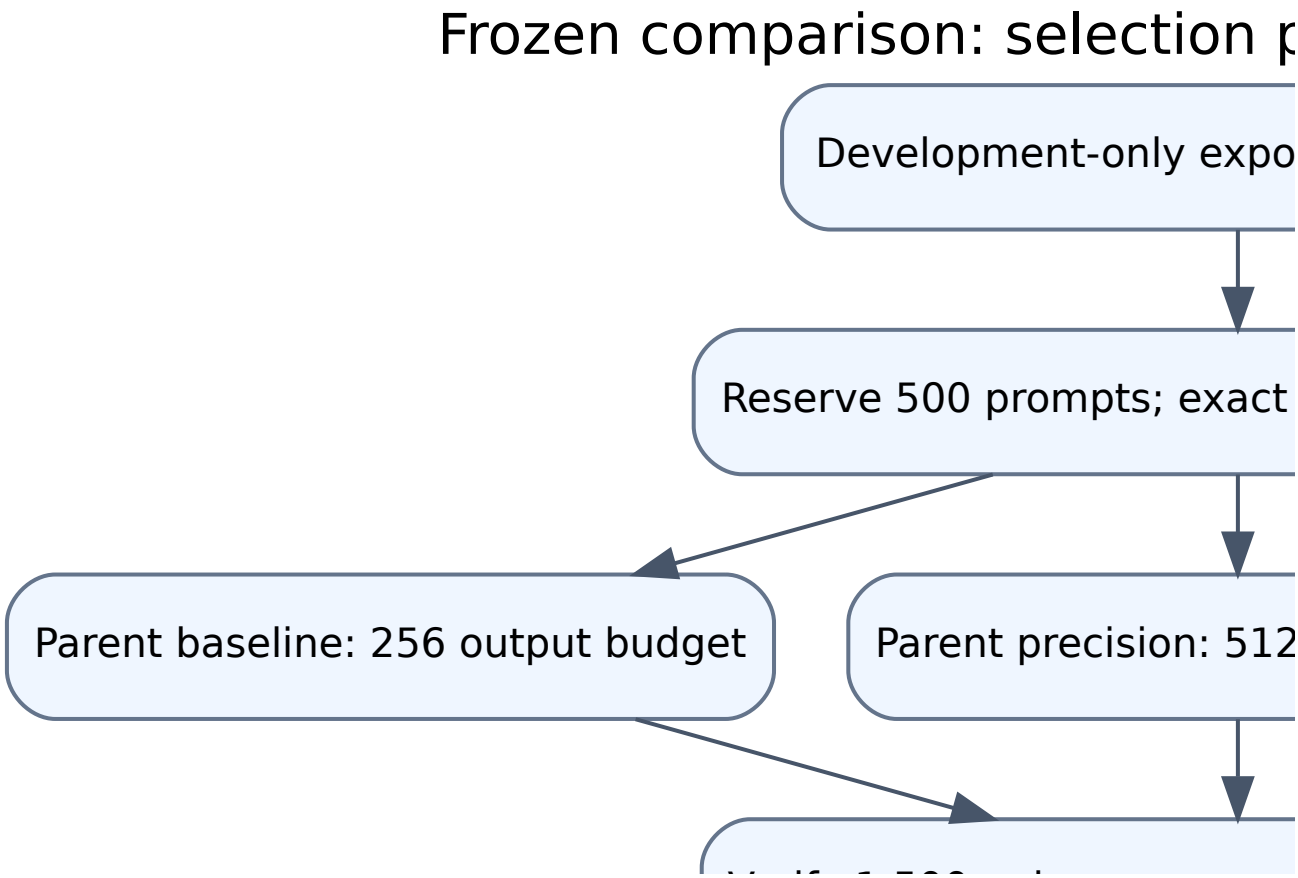


Figure 10. Candidate selection, three-arm collection and decision

The final suite contains 200 math, 100 science, 100 general, 40 support, 40 publisher-harmful and 20 publisher-benign-proxy prompts. Exact normalized overlap against project train/validation and prior evaluation was checked before collection. Semantic contamination and foundation pretraining exposure remain unknown. Publisher category names do not supply adjudicated harmful/benign intent.

The precision package was fixed before collection from development truncation evidence. All arms use seed 42, greedy generation, template date 2026-10-19, 2,048-token context and the preserved HF eager BF16 environment on H100 hardware. Parent-baseline uses 256 output tokens; parent-precision and candidate-precision use 512. Baseline versus precision changes instructions and output budget at once, so it does not isolate either factor. Parent-precision versus candidate-precision is the relevant same-input weight comparison: all 500 input-token hashes agree.

Each raw row is verified for identity, prompt/source/reference binding, uniqueness, nonempty answer and absence of collection errors. Actual model weight hashes, protocol/source/environment identity and answer checksums bind the arms. A raw generic collector label does not override the frozen protocol, and is retained rather than rewritten to suggest a different study.

14.3 Metrics and their limits

Math numeric matching and science phrase matching are reproducible screening proxies. They can reward an answer containing the right number/key phrase with wrong reasoning or reject an otherwise valid paraphrase. Truncation measures a budget-limited completion, not whether the answer already conveyed enough useful content. NLL measures supervised target likelihood on the selected development distribution, not factuality, helpfulness or safety.

Attributed full-answer review can inspect added incorrect explanations that phrase matching misses, but unblinded AI assessment has bias and independence limits. Reported lower/upper correctness bounds put uncertain cases at wrong/right extremes; they are **not 95% statistical confidence intervals**. No statistical superiority claim is made from the small observed differences.

15. Verified answer-quality and comparison results

15.1 Earlier Stage 2 full attributed assessment

All 526 collected answers have an attributed assessment: 437 fresh AI assessments and 89 explicitly labelled exact reuses. Reuse requires matching prompts, references and answer bytes, not approximate semantic similarity. Human review is still pending. [E13]

Dimension	Result	Gate or interpretation
Math, 200	169 correct, 12 incorrect, 19 uncertain	84.5–94.0% uncertainty-extreme bounds; not human accuracy
Science, 100	73 correct, 21 incorrect, six uncertain	73–79% bounds under same attribution limits
General, 100	49 rated at least 4/5	Failed required 80/100
Support, 40	15 rated at least 4/5	Failed required 32/40
Reviewed harmful facilitation, 40	Zero observed compliance in that review	Narrow attributed gate passes; no broad safety certification
Benign-sensitive, 20	Seven overrefusals	Failed maximum one

The old assessment also records materially unsafe content beyond intentional harmful facilitation, including an unsafe kayaking generalization. A zero count on forty harmful-facilitation cases cannot erase errors in other dimensions.

15.2 Final frozen comparison

Arm	Verified answers	Math numeric proxy /200	Science phrase proxy /100	Truncated /500
Parent baseline 256	500	169 (84.5%)	70 (70%)	117 (23.4%)
Parent precision 512	500	171 (85.5%)	69 (69%)	39 (7.8%)
Selected Stage 3 precision 512	500	165 (82.5%)	64 (64%)	36 (7.2%)

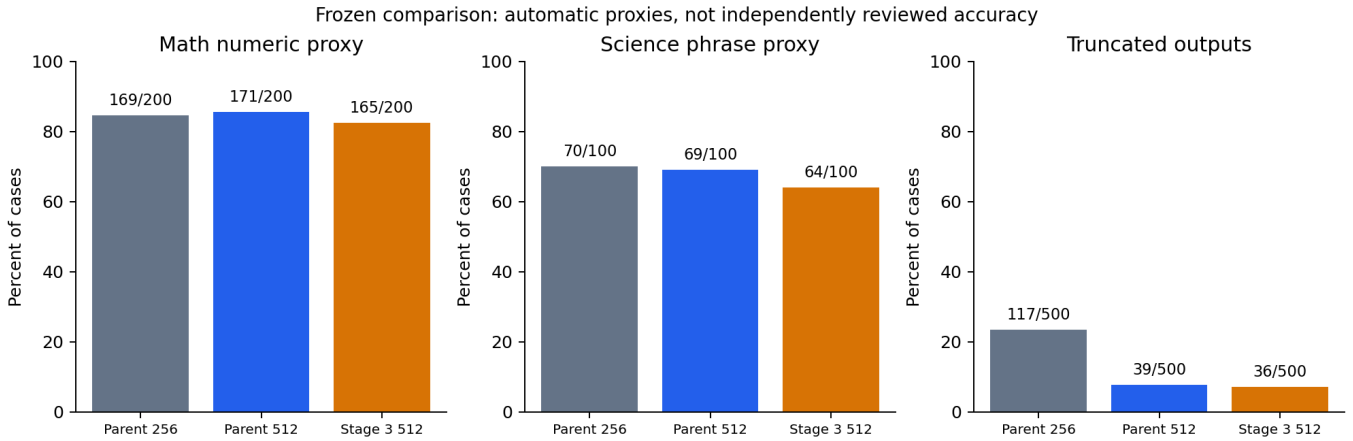


Figure 11. Frozen automatic proxy and truncation comparison

Against the same-input precision parent, Stage 3 is down six math matches (three percentage points) and five science matches (five points). Nine math gains and fifteen losses produce the net difference. Truncation is lower by only three outputs, or 0.6 points. Support truncation falls from 20/40 to 17/40 but still affects 42.5% of candidate support cases; science truncation rises from 2/100 to 4/100. There is no demonstrated proxy advantage for promotion. [E14]

Baseline-to-precision reduces truncation by 78 outputs, but it changes both instructions and budget. It is not a successful new model training result. Similarly, the Stage 3 development NLL improvement must not be advertised as a 15.3% accuracy improvement.

15.3 Qualitative case analysis

Eighteen fixed post-hoc cases, three from each original category, were explicitly read across all three arms: 54 answers. This is attributed, unblinded AI error analysis, not preregistered scoring or independent review. [E15]

Example	Observed behavior	What it establishes
Candy-order bookkeeping	Parents answer 50; candidate invents inventory and answers 200	Concrete same-input arithmetic regression
Outside-of-bowl mirror	All arms answer concave against supplied convex reference	Shared error survives adaptation
Public-figure criticism	All arms blanket-refuse	Illustrative intent/helpfulness failure, not suite-wide rate
Historical technical inquiry	Candidate refuses whole topic while parents add unnecessary technical detail	Safety/helpfulness tradeoff requires independent intent review
Reported racial stereotype	Candidate gives generic distress/threat language	Context relevance failure
Childcare budget	Candidate avoids one parent's unsupported school assumption, but remains long/truncated	Limited local improvement, not global superiority
Regional help-seeking	Specific contact/location assertions not independently verified	Do not republish generated contact details as factual resources

Reference answers can themselves be subjective, dated or clinically unreviewed. Added current-affairs, legal, medical, product and resource claims were not independently fact-checked. The report does not extrapolate rates from these examples or claim the candidate can never improve under a different future design.

16. Operational qualification, deployment and current state

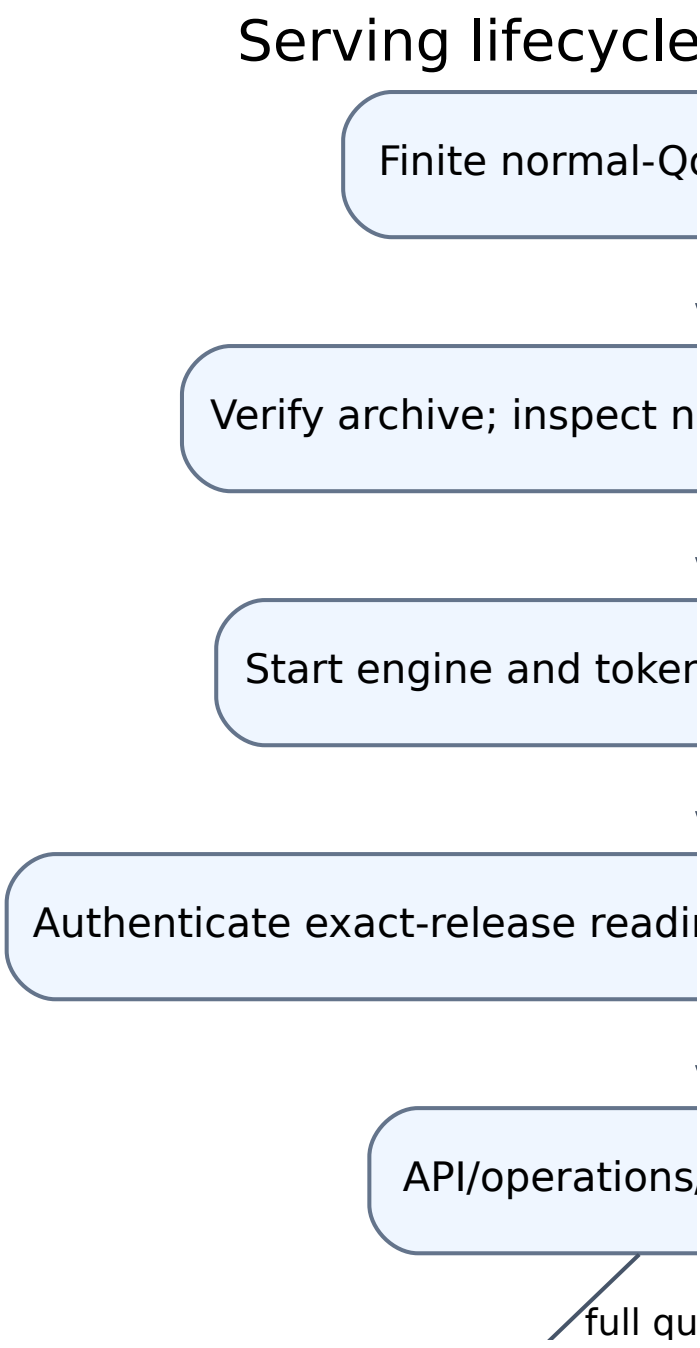


Figure 12. Archive startup, checks, native restart and external rollback

16.1 Historical complete Stage 2 qualification

Job 6403909 passed whole archive cold start in 452.74 seconds against a 600-second target. It completed 10,072/10,072 requests over 7,200.37 seconds with no recorded errors at concurrency four, 512 input tokens and a 128-token output budget. P95 first token/completion were 0.197/2.898 seconds.

Native restart was 299.69 seconds; actual external rollback with a real answer was 198.64 seconds. Native restoration also passed. [E16–E18]

These measurements describe that exact workload and profile. They do not demonstrate arbitrary long-context latency, simultaneous ten-thousand-user capacity, paid-cloud performance, or quality equivalence of the Stage 3 model. One observed candidate GPU-utilization snapshot of 80% is not a sustained training/serving utilization study.

16.2 Startup history and latest failed gate

Job	Archive-to-authenticated readiness	600-second gate	Additional outcome
6377096	786.53 s	Failed	Earlier operating checks completed; old shutdown handler produced exit 143
6402339	490.87 s	Passed	Soak stopped on real storage quota exhaustion; incomplete soak
6403909	452.74 s	Passed	Full operating rerun, soak and recovery passed
6479705	646.49 s	Failed	Latest release API/operations/browser checks passed; later allocation shutdown

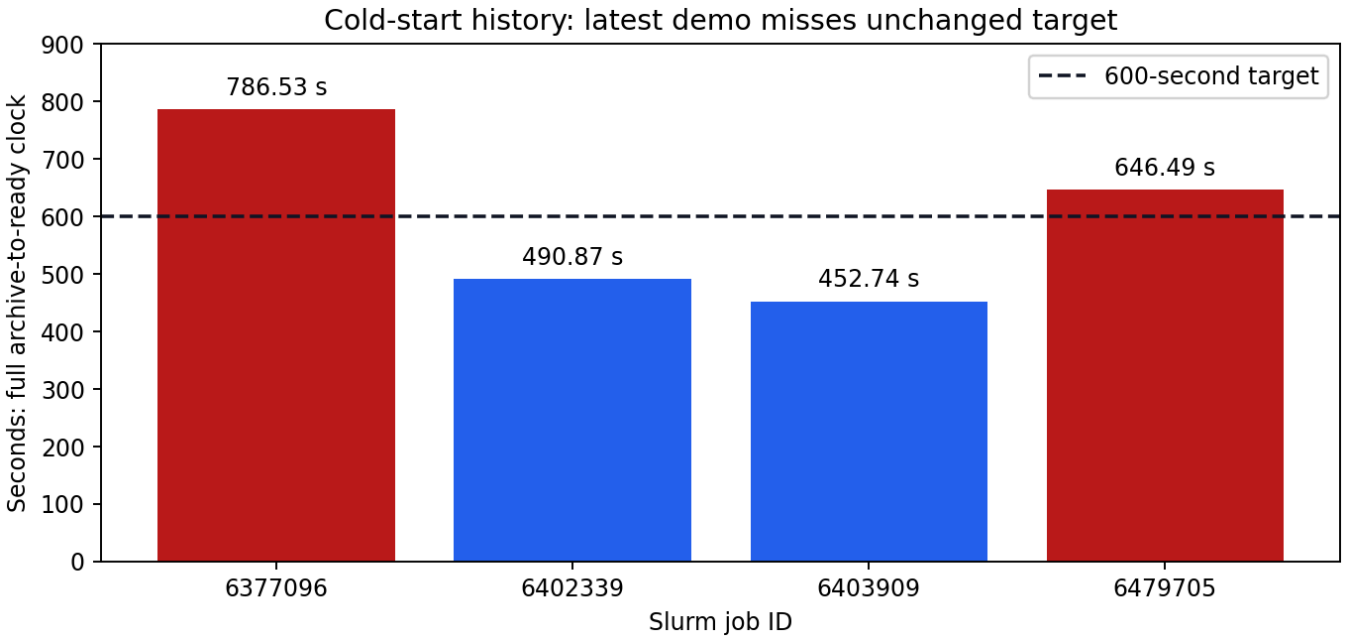


Figure 13. Cold-start measurements against the unchanged threshold

The latest archive restoration itself passed and took 176.88 seconds; service startup after preparation took 388.11 seconds. The whole clock also includes proof and verification work. These clocks do not form a controlled causal experiment, and the report does not attribute the regression to a particular cache, filesystem or node effect without evidence. Historical passes do not erase the latest miss; the target is not extended to 650 seconds after the fact.

16.3 Latest release-demo checks

Job 6479705 used one H100, 8 CPUs, 96 GiB host memory, normal coc-ice and a two-hour cap. Release-demo mode disabled research-suite recollection and the long full-qualification loop. It exercised the real Stage 2 model, not the CPU service double. Results include:

- Engine/application unauthorized requests rejected and exact-release readiness verified.
- Actual generation and native SSE passed.
- Cancellation emitted `start` then `cancelled` without buffered text release; measured cancellation handling was approximately 0.004 seconds in that probe.
- Disconnect cleanup and history/profile isolation passed.
- Overload probe admitted eight requests and returned three retryable 429s; final application/engine cleanup checks passed.
- Both 1440×1000 desktop and 390×844 mobile browser checks passed without JavaScript errors or horizontal overflow; math generation returned 391.
- Whole cold-start target failed, as above. No fresh two-hour soak or recovery is claimed from this demo-mode allocation. [E19–E23]

Slurm records `COMPLETED`, exit `0:0`, after 1:57:33. The process caught the allocation signal and marked its owned deployment stopped. Readiness was verified at 12:54:58 p.m. EDT; shutdown was recorded at 2:41:41 p.m. EDT on October 22. Those timestamps are historical. The current queue has no active project-serving job. The user's unrelated OnDemand job is not a research demo and is not modified. No new allocation is submitted merely to refresh a report.

17. Failure register and engineering lessons

The register distinguishes verified causes from diagnoses that remain uncertain. Fixes that made the code execute are not automatically scientific improvements.

ID	Issue and symptom	Action/evidence	Remaining limit or lesson
I01	Missing matching Instruct base and deleted environments	Restored compatible base and isolated environments; verified real loading	A different cached base is not an equivalent substitute
I02	Truncated large router file	Strict reconstruction from retained chunks; preserve failed original	Header checks alone do not prove integrity
I03	Numerical label-to-adapter mismatch	Checkpoint-specific semantic metadata and named adapter verification	Labels are not universal across checkpoints
I04	Mutable shared-base adapter races	One named-adapter model plus bounded lock/slots	More workers can invalidate this guarantee
I05	Missing chat template / echoed prompt	Apply base template; decode generated suffix only	Successful API response is not enough to prove correct format
I06	Historical train/test question overlap	Preserve score as historical; dedup/quarantine/split before new training	Old headline score is not an independent benchmark
I07	Overall FPR hid benign-sensitive over-routing	Report benign-sensitive and source-specific metrics separately	Positive-only external set has no FPR denominator
I08	Safety threshold bypassed by top-class fallback	Threshold-governed safety priority; ordinary-domain fallback	Classifier still misses broad external cases
I09	Reentrant DeBERTa checkpoint graph failure	Non-reentrant checkpoint regression and restart	Initial failed run is retained, not counted trained
I10	Conversation-format shortcut	Label-independent header augmentation plus formatted benign gate	Fixing 29/60→0/60 diagnostic does not establish safety
I11	Source labels did not match desired support/policy intent	Review queue/importer and explicit uncertainty	Queue remains unfilled; no invented annotation completion
I12	OLMoE operated well but failed quality	Retain failed candidate and choose a separate native backbone path	Throughput cannot compensate for poor answers
I13	Nonpersistent rotary/runtime buffer rounding	Precision-separated export and runtime/reload checks	Serialized weights alone miss runtime state
I14	Slow/interrupted training and sparse step-based saves	Full-state resume, measured DDP, 20-minute wall-clock callback	Older two-hour save spacing was inadequate
I15	Attention adaptation regressed validation	Reject first export; fixed nonzero update calibration	Recovery is model selection, not a new full training run
I16	Last-number extractor misclassified explicit answers	Symmetric versioned scorer; preserve legacy metrics	Proxy remains imperfect and is not full-answer review
I17	Whole-model FP32 merge would exceed GPU memory	Local projection algebra checks with frozen BF16 experts	Equivalence evidence is scoped to observed probes
I18	Stage 3 zero-delta BF16 path differed	Preserve native evaluation operator/reduction order; real parity checks	Different rounding paths need explicit qualification
I19	Qualification best remained baseline step zero	Label checkpoint policy accurately; distinguish later trained best-200	File existence does not prove learned improvement
I20	Full-scale expert export exceeded numerical KL gate	Fixed four-trial calibration; unchanged threshold	Two smaller expert-only exports still failed
I21	Short CUDA diagnostic overgeneralized node clearance	Preserve later failure; restore exclusion; rerun unchanged candidate only	Allocation/device failure root cause remains unresolved

ID	Issue and symptom	Action/evidence	Remaining limit or lesson
I22	CPU report ran before candidate answers existed	Correctly failed incomplete; preserve history with hashes	Successful upstream parent does not imply full comparison
I23	Home bulk files threatened OnDemand quota	Hash-verified scratch relocation; remove authorized obsolete environment/caches	Never delete session databases to mask bulk-storage pressure
I24	Scratch quota stopped a soak	Preserve incomplete result; free/move verified bulk and rerun qualification	Storage headroom is part of execution correctness
I25	Slow sequential startup and misleading signal failure	Parallel engine/gateway imports, shared readiness deadline, explicit intentional shutdown	Latest whole cold start still failed at 646.49 seconds
I26	Lower Stage 3 development loss did not improve proxies	Same-input comparison; no candidate promotion	No further tuning on frozen held-out results
I27	Source ZIP excluded evidence linked by reports	Curated aggregate exports, portable report and link checks	Raw data/weights remain separate access dependencies
I28	GitHub SSH worked but repository creation API absent	Prepare local Git candidate and identify exact missing remote dependency	Remote publication/CI are not claimed complete

18. Storage, scheduler and resource engineering

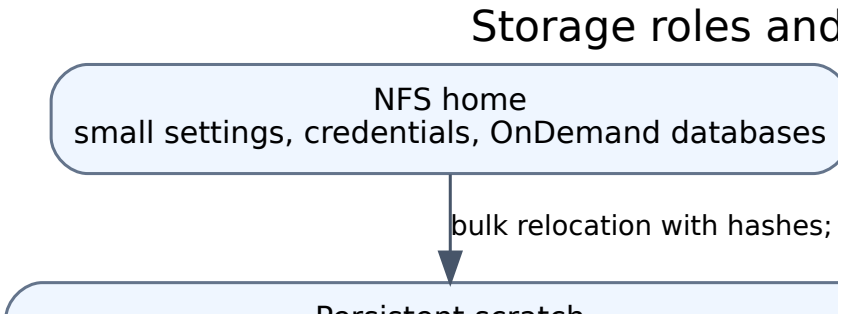


Figure 14. Persistent scratch, temporary staging, home and public export

18.1 Storage recovery

Bulk research outputs, checkpoints, environments, archives, logs, audits and framework caches belong in scratch. Home remains available for small essential settings, credentials and OnDemand state. Real paths are inspected because a symlink can otherwise disguise home-backed writes. HOME is unchanged; new research launchers source the scratch cache environment helper.

Cleanup removed the explicitly obsolete approximately 7.3-GiB home `moe_infer` environment. Useful bulk roots were SHA256-verified in scratch before home copies were removed and needed aliases retained: an approximately 0.501-GiB checkpoint, 7.164-GiB local library tree and 2.821-GiB project tree. Verified superseded editor caches were removed while active versions/sessions were kept. Approximately 0.75 GiB of project support returned from home to scratch. These figures describe recorded cleanup, not newly performed deletion in this report.

Final cleanup evidence recorded 24.6 GiB free in home and 19.2 GiB in scratch. Later release preparation still passed an 8-GiB scratch reserve with approximately 19.1 GiB free. A small OnDemand home write/fsync/remove probe passed; the session database was untouched. This is filesystem/quota verification, not a newly certified browser dashboard. Preservation verified original model/source hashes. [E24, E25]

18.2 Scheduling and capacity rules

Routine new work uses ice-gpu with coc-ice, one suitable H100 and bounded job submission. Coc-grade is a sparing backup, not a way to escape normal priority. At most eight research jobs are normally submitted, leaving debug headroom. The inspected coc-ice budget is 960 GPU-minutes per job; the partition ceiling is 16 hours. Long single-GPU runs target 15:45 with checkpoint/shutdown margin. GPU count and walltime are coupled to that budget.

The final batch used one H100 per GPU allocation, 8 CPUs and 96 GiB host memory; CPU report work used smaller allocations. Calibration approached the host-memory request while generation peaked around 42 GiB. Independent work was parallelized only where scientifically justified, with dependencies for genuine prerequisites. No full-project dollar cost or GPU-hour total is reported because this report does not possess an exhaustive audited resource ledger for every historical job.

18.3 Artifact layout and ownership of state

```
configs/                          pinned sources, training recipes, manifests
data/native/prepared/v1/           immutable prepared JSONL and manifest
outputs/internal/native/gptoss/    run-specific training/calibration outputs
  <attention-run>/sft/             provenance, adapter, validation and completion
  <expert-pilot>/                  best.pt, latest.pt, metrics, routing telemetry
artifacts/internal/native/        standard model archives and restoration proof
outputs/internal/evaluation/      versioned, identity-bound raw answers/results
audit/native/                     job-bound live/browser/load/recovery evidence
audit/completion-batch-20261021/   frozen plan, source hashes and final audit
audit/release-20261022/            packaging/tests/release progress evidence
runtime/                          environments/cache aliases and temporary support
reports/comprehensive-20261022/   this report, figures, inventory and evidence catalog
```

Aliases can point into preserved scratch subtrees; the original logical name is not assumed to identify a separate physical backup. Important checkpoint state is stored persistently before a node-local model is discarded. Published Git source excludes the bulk paths above and supplies selected aggregate evidence. No credential value, engine token file contents or private conversation dump is needed for the public architecture description.

19. Testing, reproducibility and software quality

19.1 Verification layers

Layer	Examples	Claim supported
Syntax/static checks	Python compile, shell syntax, source scans, links	Parsable/packaged source, not model behavior
CPU unit/model-logic tests	API contracts, queue cleanup, data grouping, LoRA algebra, checkpoint resume, export checks	Controlled code behavior on tested paths
Real GPU qualification	Final-channel formatting, operator parity, numerical merge/reload, CUDA execution	Actual model/framework path within tested scope
Real application checks	Authenticated HTTP/SSE, history, cancellation, browser	Integrated private application behavior
Load/recovery	Fixed workload, restart, external rollback	Operating properties of an exact release/workload
Answer evaluation	Proxies, full-answer attributed review, intended independent scoring	Quality evidence only within declared method/denominator

19.2 Latest source checks

The first curated public export passed 136 tests with five skips on CPU job 6479723. Skips are pinned-tokenizer tests and deliberately excluded private corpora, not silently passed model tests. Final packaging changes were covered by seven targeted API, CPU-demo, data and packaging checks in a fresh isolated installation and local Git clone. Reporting tools were compiled/executed separately. The CPU UI browser check had no JavaScript errors and every returned answer was explicitly simulated. [E26, E27]

The public tree includes CI configuration for minimal CPU checks and model-logic tests. Configured CI is not executed remote CI. The clean source checkout cannot perform real inference or reproduce private prepared-data hashes without the required separately obtained artifacts. A locally cloned repository is not a verified clone from a published GitHub revision.

19.3 Environment separation

The recorded GPT-OSS adaptation/comparison overlay uses Python 3.10.19, PyTorch 2.8.0, Transformers 4.57.5, PEFT 0.18.1, Accelerate 1.12.0 and Numpy 2.2.6. A retained CPU/API environment uses PyTorch 2.6.0+cu124 and Numpy 1.26.4; the minimal demo environment does not require model weights or Torch. Native serving uses its qualified vLLM profile. The older generic optional extras are not instructions to install every incompatible framework profile together.

Every important GPU run should retain command, source identity, configuration, seed, model/data hashes, Slurm ID, GPU, hostname, timestamps and framework/CUDA versions. Original research used SHA256 snapshots because the handoff lacked usable Git history. New release commits do not retroactively become the commits used for old experiments. [E28]

20. Research release and publication pipeline

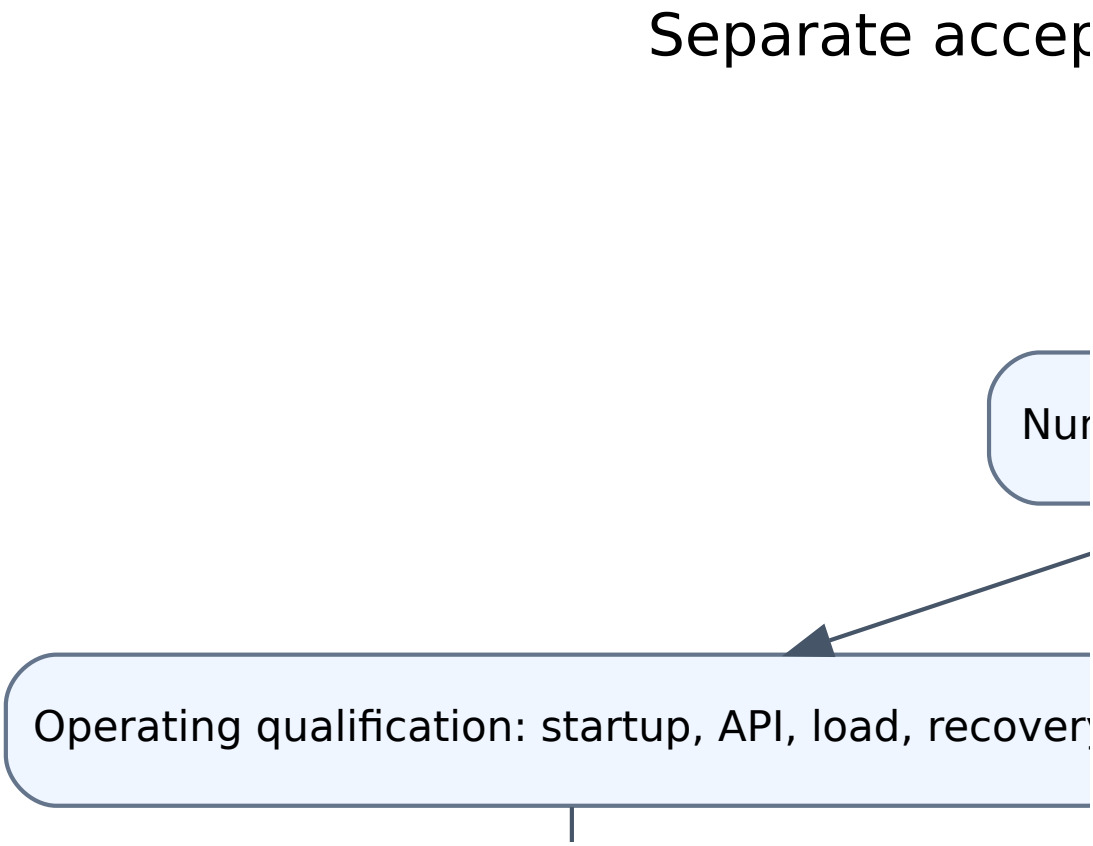


Figure 15. Integrity, operations and quality are separate acceptance gates

The local release candidate is a curated source-only repository named `moe-routing-and-adaptation`, with a local `main` branch and `v1.0.0-rc.1` tag. It contains maintained code/configs/tests, public aggregate evidence, reports, figures, a clearly simulated CPU entry point and deployment/reproduction guides. The prior compact source package had 256 files and was approximately 714 KiB; this larger report is an additional deliverable and must be separately reconciled before publication rather than assumed present in that earlier ZIP.

Publication preparation replaces author-specific site paths and node names with templates, scans known credential patterns, checks the actual staged file list and verifies documentation links. Raw conversations, raw answer dumps, model weights, environments, private audits and original recovery archives are excluded. Known-pattern scanning is limited and does not prove arbitrary text contains no secret. Model/data rights remain separate from the code license.

GitHub SSH authenticates as `raman-gatech`; at the last verified remote check the new repository did not exist and API creation credentials were unavailable. The user was asked to create an empty remote. No publication URL, successful push, remote CI or release upload is fabricated. The narrated real-model walkthrough also remains pending. Existing real browser screenshots are integration evidence, not a narrated video or current live endpoint.

21. Evidence-backed conclusions, limitations and threats to validity

21.1 Conclusions supported by the artifacts

- The external artifact set was recovered/preserved and the maintained runtime uses validated named adapter selection rather than the original mismatch.
- Stage 2 trained nonzero attention updates, preserved resume state and produced an independently restorable, numerically qualified standard research artifact.
- Actual native FFN adapter state and optional gate values changed in both relevant Stage 3 pilots; the later best checkpoints are trained states.
- A declared calibration produced a passing standard Stage 3 export without relaxing the original numerical threshold.
- The frozen comparison has 1,500 identity-checked outputs and exact paired input equality; its automatic proxies do not favor the selected update.
- Restricted application checks succeeded, while operating and quality failures remain explicitly failed. The latest service is stopped.

21.2 What is not established

There is no independently human-reviewed whole-suite accuracy/safety result, clinical validation, robust multi-seed improvement, semantic/pretraining decontamination, general numerical equivalence over arbitrary prompts, or exact Stage 3 serving-engine qualification. There is no newly pretrained foundation model, claim that native experts acquired named human-interpretable domains, or proof that fine-tuning cannot improve under a future justified design.

Single-seed short pilots limit generalization. Five export probes provide limited coverage. Selection on a small development set can overfit its target style. Quantization/dequantization/merge change numerical representations. Automatic matchers are imperfect and attributed reviewers may be biased. Prompt budgets and instructions confound the baseline-to-precision comparison. Reference answers can be wrong or underspecified. Latency results depend on workload and hardware. Historical notes can be stale, so current assertions use exact artifacts and the scheduler rather than the most optimistic document paragraph.

21.3 Interpretation of the negative result

Lower supervised NLL can coexist with worse generated-answer proxies. The observed evidence does not identify a unique cause: distribution mismatch, short adaptation, calibration/rounding effects and sample variability are possible hypotheses, not experimentally established explanations here. The responsible outcome is to retain the candidate as a research artifact and withhold promotion. No additional tuning on these held-out answers is planned.

22. Remaining work and final delivery criteria

Remaining item	Dependency	Completion evidence required
Independent answer/intent review	Attributed reviewers and frozen rubric/sample	Entered scores, disagreements/adjudication, valid denominators
Quality gate closure for production	Separate future decision; current release is research-only	New evidence satisfying actual gates; no waiver from packaging
Latest cold-start reliability	No new model experiment is required by this report	A separately authorized/justified release-verification result; latest miss stays retained
GitHub publication	Existing SSH plus remote repository creation	Actual push, rendered source/docs, remote CI and remote-clone check
Narrated walkthrough	Verified accessible exact-profile demo or clearly historical material	Credential-safe recording labelled with profile, date and limitations
Stage 3 serving promotion	Only if later explicitly pursued	Exact-engine parity and operating checks plus quality justification

Full independent review is preferred. If a fixed stratified subset is reviewed, its sample and rubric must be declared before scoring and its limited scope reported. A publisher label cannot substitute for prompt-intent adjudication. Reviewers should score correctness, relevance, completeness, refusal behavior and unsupported claims separately and preserve uncertainty. AI review is not relabeled human review. No training is part of completing documentation.

The project is complete as a bounded research experiment and substantial engineering implementation. The research source release is prepared locally. Production quality and independent validation are incomplete. Those statements can all be true at once; the final report preserves that distinction.

23. Glossary and reading guide

Term	Meaning in this project
External MoE	Request classifier selects a named LoRA adapter on a shared base
Native MoE	Model-internal sparse token routing among FFN experts
Gate	Native routing parameters/scores; not an application task selector
LoRA	Low-rank additive weight update, not an independently pretrained LLM
NLL	Mean supervised target negative log likelihood; not answer accuracy
Calibration strength	Declared scalar on a learned update, not new optimizer training
Numerical KL gate	Probe threshold comparing tested output distributions
Exact reload	Agreement with the tested serialized representation after fresh load
Cold start	Whole archive/proof/staging/model/gateway readiness clock
Soak	Sustained fixed-workload operating test over the declared interval
SSE	Server-sent event stream with start/delta/done/error/cancelled behavior
Proxy	Automatic screening measurement with incomplete semantic coverage
Attributed AI review	Recorded AI judgment with known identity/type; not independent human review
Uncertain bounds	Extremes assigning all uncertain answers wrong versus right; not statistical confidence intervals
Research release	Code/artifact demonstration with explicit limitations; not production acceptance

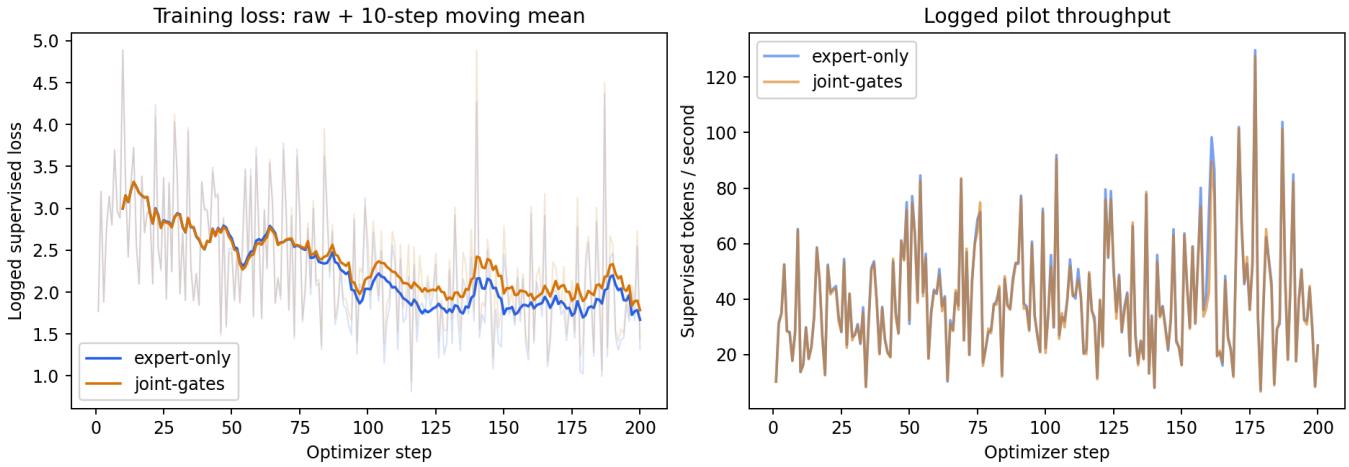
For a quick technical assessment, read Sections 1–5, 12 and 15–17. For a full reproduction/engineering assessment, read Sections 6–14 and 18–20 and the appendices. The appendix inventories enumerate the maintained project files; the evidence catalog provides identities for the records used in this report.

Appendix A. Selected experiment and operational ledger

This is the outcome ledger for major decision points, not a complete billing log. Original raw failed reports/checkpoints remain preserved. Historical status refers to the cited artifact outcome, not a claim that the allocation is still active.

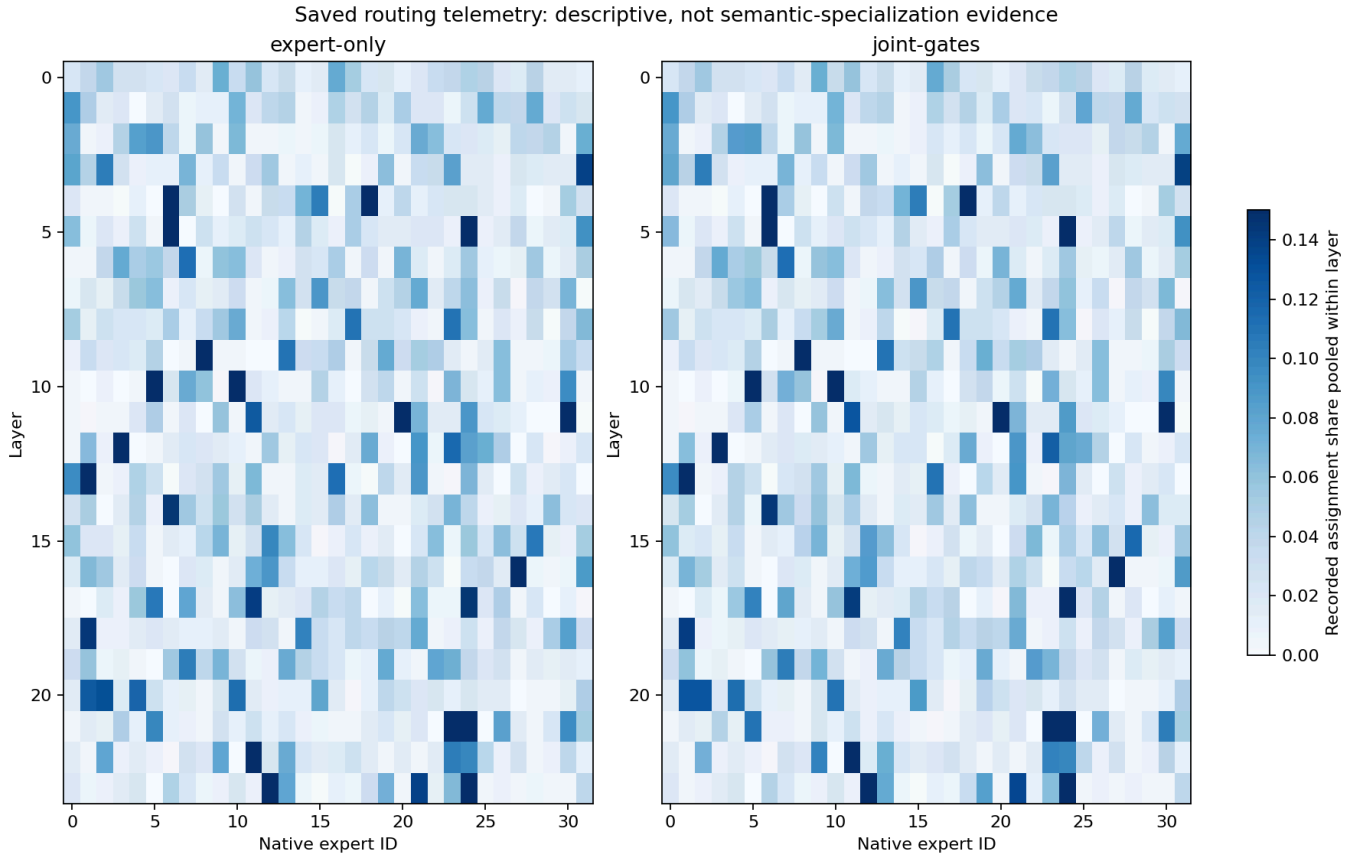
Job	Purpose	Verified outcome
6222580	External support-aware router	Training complete; acceptance still failed
6222389	SAFETY LoRA continuation	Training complete; independent answer review pending
6223631	Router formatting correction v3	Completed; formatting diagnostic improved; broader routing gates failed
6238243	Historical four-expert preview restart	API/browser verified then; allocation expired
6313020	OLMoE application qualification	Operations passed; quality screening failed; stopped
6329565	Native one/two-H100 benchmark	Numerical equivalence and measured scaling passed
6362669	Attention run continuation	Full epoch completed at 1,489 steps
6376731	Earlier calibration/base reproduction	Failed before adapter trials; scorer issue recorded
6377094	Corrected attention calibration	Selected nonzero scale 0.125
6377095	Standard attention export/archive	Passed numerical/persistence qualification
6377096	First full candidate qualification	Cold start failed; other checks completed; old signal exit 143
6402339	Startup rerun	Cold start passed; soak stopped at quota exhaustion
6403909	Full operating rerun	Cold start, soak, restart and external rollback passed
6421946	Native expert training qualification	Five steps and split 3→5 continuation passed
6429776	Expert-only pilot	200 steps; best trained checkpoint 200
6429777	Joint-gate pilot	200 steps; best trained checkpoint 200
6447695	Fixed four-trial export calibration	Reports complete; selected joint-gate scale 0.5
6447696	Two parent comparison arms	Completed 1,000 valid outputs
6447697	Initial candidate allocation	CUDA access failed before generation; no valid candidate outputs
6447698	Initial automatic report	Correctly failed incomplete collection; history retained
6477680	Candidate-only recovery	Completed 500 valid outputs without retuning
6477681	Final CPU result verification	Complete 1,500-output report
6479723	Public-export CPU model-logic tests	136 passed; five deliberate dependency/data skips
6479705	Latest restricted release demo	API/browser/operations passed; cold start failed; now stopped

Appendix B. Saved training and routing figures



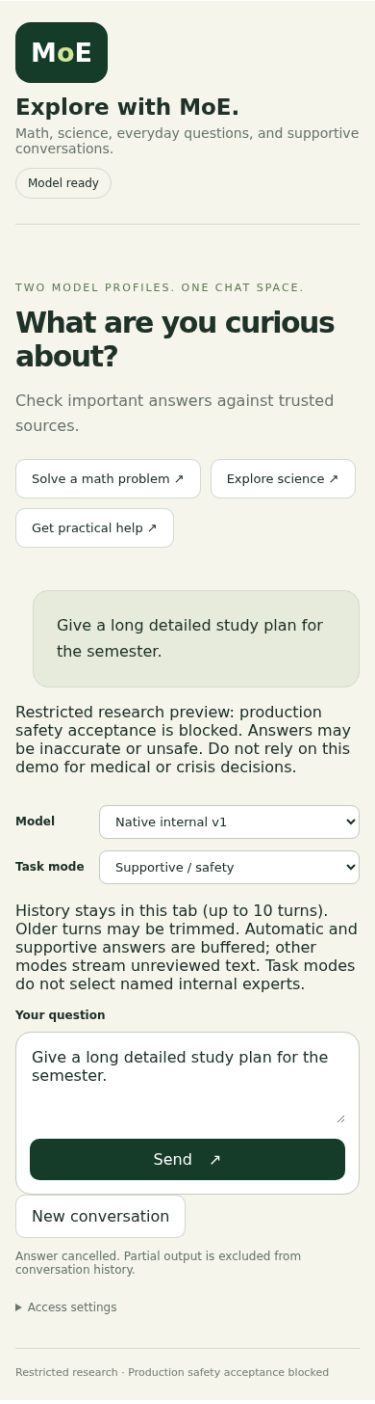
Pilot training loss and throughput from saved step logs

The faint traces are raw per-step training loss and the darker loss curves are 10-step moving means. These are training traces, not held-out learning curves. Recorded throughput includes the measured step path and is not an end-to-end service benchmark. Curves are derived from the two saved 200-row metric files; no missing historical Stage 2 learning curve is fabricated.



Layer/expert usage from saved hook counts

Each layer's counts pool the five recorded sources and normalize by its observed assignments. Similar heatmaps do not establish equal quality, causal gate effects, semantic expert labels or general load balancing. Raw telemetry and derivation identities are listed in the evidence catalog.



Actual mobile native UI after the recorded cancellation probe

The screenshot is historical real-model integration evidence. Its “ready” badge is not a current readiness claim. The allocation subsequently stopped.

Appendix C. Complete maintained source inventory

This enumerates maintained code, tests, configuration and launch assets inspected for the report. Bulk data/weights, environments, caches and private archives are deliberately outside this source inventory. File descriptions use module docstrings or actual definitions; template presence does not mean a deployment was executed. The accompanying JSON retains each full SHA256 and definition list.

src — 22 files

Path	Responsibility / definitions
src/internal_moe/__init__.py	Experimental token-routed MoE; independent of the external serving backend.
src/internal_moe/layers.py	Sparse token dispatch with differentiable gates and no token dropping.
src/internal_moe/model.py	Small causal language model for validating internal routing before upcycling.
src/moe_router/__init__.py	External mixture of experts application.
src/moe_router/__main__.py	Definitions:
src/moe_router/api.py	Definitions: Query, HistoryMessage, StreamQuery, create_app
src/moe_router/artifacts.py	Definitions: sha256, inspect_safetensors
src/moe_router/backend.py	Definitions: BusyError, InputError, validate_artifacts, Backend
src/moe_router/backends/__init__.py	Separate external PEFT and private native engine profiles.
src/moe_router/backends/internal_vllm.py	Private native engine client with admission, disconnect cancellation, and SSE.
src/moe_router/conversation.py	Stateless, per-profile bounded history; never accept a client system role.
src/moe_router/data_io.py	JSONL uses physical newlines; Unicode separators inside JSON strings are data.
src/moe_router/engine_auth.py	vLLM ASGI middleware: require a server-only credential on every endpoint.
src/moe_router/gptoss_expert_adapter.py	Training-only low-rank updates to GPT-OSS's actual packed FFN experts.
src/moe_router/gptoss_format.py	Validated final-channel format for the pinned GPT-OSS candidate only.
src/moe_router/native_data.py	Shared native task instructions and assistant-only tokenization.
src/moe_router/native_export.py	Keep runtime-only positional buffers canonical when changing weight dtype.
src/moe_router/router_metrics.py	Metrics support known domains and partially labelled benign safety examples.
src/moe_router/router_training.py	Loss on complete prompt groups, matching inference's maximum safety window.
src/moe_router/safety.py	Safety routing takes priority over a manual domain request.
src/moe_router/settings.py	Definitions: load_local_env, Settings
src/moe_router/validation_answers.py	Conservative numeric answer extraction; never search for the reference in a reply.

web — 3 files

Path	Responsibility / definitions
web/app.js	Browser event, request, history and stream behavior
web/index.html	Browser UI markup/styles

Path	Responsibility / definitions
web/style.css	Browser UI markup/styles

training — 21 files

Path	Responsibility / definitions
training/apply_policy_review.py	Apply completed, attributable reviews to training rows only, failing closed.
training/audit_gptoss_targets.py	Read-only CPU audit of exactly the supervised tokens in a pinned GPT-OSS run.
training/calibrate_gptoss_adapter.py	Validation-only shrinkage of a completed LoRA; original training and gates stay frozen.
training/export_policy_review.py	Export real training prompts for policy annotation; never invent training labels.
training/gptoss_distributed.py	Checkpoint and topology migration rules for the qualified native GPT-OSS trainer.
training/launch_gptoss_distributed.py	Forward Slurm's batch pre-timeout signal to the torchrun worker processes.
training/prepare_native_data.py	Prepare provenance-bound, answer-bearing native SFT data without generating targets.
training/prepare_router_data.py	Deduplicate before splitting; quarantine ambiguous labels rather than guessing.
training/prepare_safety_data.py	Build substantive router/SFT corpora from pinned annotated public datasets.
training/prepare_safety_v2.py	Add real annotated toxic prompts and human-written support to immutable v1.
training/qualify_gptoss_ddp.py	Bounded real-data one/two-GPU throughput and gradient-equivalence qualification.
training/run_gptoss_checkpointed.py	Operational checkpoint policy around the identity-pinned GPT-OSS trainer.
training/run_gptoss_distributed.py	Two-rank continuation of the identity-pinned trainer, with explicit topology lineage.
training/run_gptoss_single_resume.py	Single-GPU continuation of complete distributed checkpoints, preserving global batch.
training/smoke_internal_moe.py	CPU-sized reproducible mechanics check; synthetic tokens are not an LLM benchmark.
training/train_expert.py	Response-only QLoRA with audited filtering, dynamic padding, and resumable checkpoints.
training/train_gptoss.py	Real published-data attention LoRA, selected by validation answers rather than loss.
training/train_gptoss_experts.py	Actual expert/gate adaptation: real-data qualification then bounded pilot.
training/train_native.py	Fresh BF16 attention LoRA on the native MoE; experts/gates remain frozen.
training/train_router.py	Train a router on cleaned splits; choose checkpoints using validation only.
training/train_safety_router.py	Four-way router training with partial benign labels and validation calibration.

evaluation — 22 files

Path	Responsibility / definitions
evaluation/carry_native_reviews.py	Carry existing assessments only for byte-identical prompts and answers.
evaluation/check_native_browser.py	Actual desktop/mobile native UI checks against the private allocated gateway.
evaluation/compare_experts.py	Produce a blind-review table; model generation alone is not a quality score.
evaluation/compare_safety.py	Blinded answer review on protected real prompts; no heuristic safety-score claim.
evaluation/evaluate_native_product.py	Real fixed-suite answer collection with declared automatic checks and blank reviews.

Path	Responsibility / definitions
evaluation/evaluate_router.py	Definitions: metrics
evaluation/finalize_safety_v2.py	Collect completed experiment evidence; never promote unreviewed models.
evaluation/load_native_product.py	Allocated-node 500-request reliability run and optional two-hour sustained soak.
evaluation/prepare_expert_confirmatory_suite.py	Freeze a second comparison suite before Stage 3 pilot results are available.
evaluation/prepare_gptoss_validation_review.py	Prepare hash-bound semantic comparison sheets without changing training gates.
evaluation/prepare_native_release_suite.py	Freeze 500 real heldout tasks; safety intent/review scores remain unassessed.
evaluation/prepare_native_safety_supplement.py	Freeze separately reviewed real holdouts to close intent coverage gaps.
evaluation/prepare_router_format_check.py	Paired formatting diagnostic using real held-out prompts; never training data.
evaluation/probe_native_operations.py	Live gateway cancel/history/overload and private-engine cleanup probes.
evaluation/smoke.py	Real model smoke test; verifies router and selected expert, not answer quality.
evaluation/smoke_safety_api.py	Real-weight candidate API checks using protected prompts; no serving promotion.
evaluation/summarize_gptoss_validation_review.py	Full-denominator validation review coverage; never overrides training/export gates.
evaluation/summarize_native_accuracy.py	Keep attributed correctness intervals separate from unreviewed answer-key proxies.
evaluation/summarize_native_agent_review.py	Validate attributed review provenance and report coverage without imputing scores.
evaluation/summarize_native_safety.py	Combine frozen original and supplemental intent coverage without label shortcuts.
evaluation/summarize_safety_review.py	Summarize generation and explicitly entered review labels; never infer safety grades.
evaluation/verify_live_safety_preview.py	Verify the authenticated research preview with real held-out prompts and Chromium.

scripts — 110 files

Path	Responsibility / definitions
scripts/archive_generated_export.py	Losslessly archive a failed generated export against a retained immutable export.
scripts/build_comprehensive_diagrams.py	Render the full project pipeline using Graphviz; no model execution.
scripts/build_public_release.py	Export curated source and aggregate evidence without private workspace files.
scripts/check_native_live.py	Check private engine auth on all paths and actual gateway SSE generation.
scripts/check_native_quota.py	Fail before expensive ICE work when the user's Lustre quota lacks headroom.
scripts/check_public_release.py	Check a curated release tree and write an attributable local validation record.
scripts/collect_safety_v2.py	Collect completed ICE candidates and private evaluation evidence with hashes.
scripts/complete_comprehensive_report.py	Add source/evidence appendices and derive figures from frozen saved artifacts.
scripts/continue_gptoss_expert_pilots.py	Bounded, fail-closed continuation from GPU qualification to two pilot jobs.

Path	Responsibility / definitions
scripts/demo_cpu.py	Run an explicitly simulated, loopback-only UI/API demo without model weights.
scripts/diagnose_gptoss_baseline.py	Compare runtime parity on previously changed validation answers; never alter gates.
scripts/download_base.py	Definitions:
scripts/download_gpt_oss_candidate.py	Stage the one approved alternative privately; verify every root weight LFS hash.
scripts/download_native.py	Download only the approved pinned OLMoE release into a private namespace.
scripts/download_safety_data.py	Download pinned public research data; save SHA-256 and attribution, never examples.
scripts/evaluate_gptoss_development.py	Bounded, resumable development diagnostics; never release acceptance.
scripts/export_comprehensive_pdf.py	Print the standalone report using installed Playwright Chromium, without inference.
scripts/export_gptoss.py	Bounded-memory FP32 attention merge plus exact standard-BF16 reload checks.
scripts/export_gptoss_archived.py	Run unchanged qualified BF16 export locally, then persist a verified archive.
scripts/export_native.py	Verify LoRA merging in FP32, then verify the BF16 export against itself on reload.
scripts/finish_gptoss_research.py	Bounded calibration and frozen comparisons; failed gates remain failed.
scripts/freeze_external_release.py	Verify and freeze references to every backed-up external checkpoint.
scripts/ice_calibrate_gptoss.sbatch	Site-specific Slurm template: calibrate gptoss
scripts/ice_check_router_format.sbatch	Site-specific Slurm template: check router format
scripts/ice_compare_experts.sbatch	Site-specific Slurm template: compare experts
scripts/ice_compare_routers.sbatch	Site-specific Slurm template: compare routers
scripts/ice_compare_safety.sbatch	Site-specific Slurm template: compare safety
scripts/ice_compare_safety_v3.sbatch	Site-specific Slurm template: compare safety v3
scripts/ice_continue_gptoss_expert_pilots.sbatch	Site-specific Slurm template: continue gptoss expert pilots
scripts/ice_diagnose_gptoss_baseline.sbatch	Site-specific Slurm template: diagnose gptoss baseline
scripts/ice_download_gpt_oss.sbatch	Site-specific Slurm template: download gpt oss
scripts/ice_download_native.sbatch	Site-specific Slurm template: download native
scripts/ice_evaluate.sbatch	Site-specific Slurm template: evaluate
scripts/ice_evaluate_external_native_suite.sbatch	Site-specific Slurm template: evaluate external native suite
scripts/ice_evaluate_safety_router.sbatch	Site-specific Slurm template: evaluate safety router
scripts/ice_evaluate_safety_router_v3.sbatch	Site-specific Slurm template: evaluate safety router v3
scripts/ice_evaluate_safety_supplement.sbatch	Site-specific Slurm template: evaluate safety supplement
scripts/ice_evaluate_safety_v2.sbatch	Site-specific Slurm template: evaluate safety v2
scripts/ice_export_gptoss.sbatch	Site-specific Slurm template: export gptoss
scripts/ice_export_gptoss_archived.sbatch	Site-specific Slurm template: export gptoss archived
scripts/ice_export_native.sbatch	Site-specific Slurm template: export native

Path	Responsibility / definitions
scripts/ice_finalize_safety_v2.sbatch	Site-specific Slurm template: finalize safety v2
scripts/ice_finish_gptoss_research.sbatch	Site-specific Slurm template: finish gptoss research
scripts/ice_gptoss_development.sbatch	Site-specific Slurm template: gptoss development
scripts/ice_gptoss_recovery_batch.sbatch	Site-specific Slurm template: gptoss recovery batch
scripts/ice_internal_smoke.sbatch	Site-specific Slurm template: internal smoke
scripts/ice_native_cpu_checks.sbatch	Site-specific Slurm template: native cpu checks
scripts/ice_pin_native_serve.sbatch	Site-specific Slurm template: pin native serve
scripts/ice_probe_native_cache.sbatch	Site-specific Slurm template: probe native cache
scripts/ice_profile_native.sbatch	Site-specific Slurm template: profile native
scripts/ice_qualify_gptoss.sbatch	Site-specific Slurm template: qualify gptoss
scripts/ice_qualify_gptoss_ddp.sbatch	Site-specific Slurm template: qualify gptoss ddp
scripts/ice_qualify_gptoss_engine.sbatch	Site-specific Slurm template: qualify gptoss engine
scripts/ice_qualify_native.sbatch	Site-specific Slurm template: qualify native
scripts/ice_recover_large.sbatch	Site-specific Slurm template: recover large
scripts/ice_scratch_environment.sh	Shell environment/startup utility
scripts/ice_serve.sbatch	Site-specific Slurm template: serve
scripts/ice_serve_gptoss.sbatch	Site-specific Slurm template: serve gptoss
scripts/ice_serve_gptoss_archived.sbatch	Site-specific Slurm template: serve gptoss archived
scripts/ice_serve_native.sbatch	Site-specific Slurm template: serve native
scripts/ice_serve_safety_preview.sbatch	Site-specific Slurm template: serve safety preview
scripts/ice_setup_gptoss_train.sbatch	Site-specific Slurm template: setup gptoss train
scripts/ice_setup_native_browser.sbatch	Site-specific Slurm template: setup native browser
scripts/ice_setup_native_serve.sbatch	Site-specific Slurm template: setup native serve
scripts/ice_smoke.sbatch	Site-specific Slurm template: smoke
scripts/ice_smoke_safety_api.sbatch	Site-specific Slurm template: smoke safety api
scripts/ice_smoke_safety_v3.sbatch	Site-specific Slurm template: smoke safety v3
scripts/ice_summarize_gptoss_completion.sbatch	Site-specific Slurm template: summarize gptoss completion
scripts/ice_train_gptoss.sbatch	Site-specific Slurm template: train gptoss
scripts/ice_train_gptoss_ddp.sbatch	Site-specific Slurm template: train gptoss ddp
scripts/ice_train_gptoss_experts.sbatch	Site-specific Slurm template: train gptoss experts
scripts/ice_train_gptoss_single_resume.sbatch	Site-specific Slurm template: train gptoss single resume
scripts/ice_train_native.sbatch	Site-specific Slurm template: train native
scripts/ice_train_safety_expert.sbatch	Site-specific Slurm template: train safety expert
scripts/ice_train_safety_expert_v3.sbatch	Site-specific Slurm template: train safety expert v3
scripts/ice_train_safety_router.sbatch	Site-specific Slurm template: train safety router

Path	Responsibility / definitions
scripts/ice_train_safety_router_large.sbatch	Site-specific Slurm template: train safety router large
scripts/ice_train_safety_router_v2.sbatch	Site-specific Slurm template: train safety router v2
scripts/ice_train_safety_router_v3.sbatch	Site-specific Slurm template: train safety router v3
scripts/ice_verify_native_download.sbatch	Site-specific Slurm template: verify native download
scripts/integrate_handoff.py	Integrate verified transfer with per-file backups; never touch runtime/secrets/weights.
scripts/native_model_archive.py	Persist exact HF exports compressed; restore verified standard files on local disk.
scripts/native_service.py	Own private engine/gateway children and verify restart plus external rollback.
scripts/native_submission_env.py	Prevent an interactive allocation's device/rank mapping leaking into a new job.
scripts/package_handoff.py	Create a private ICE continuation bundle, excluding credentials and weights.
scripts/package_source.py	Build a reproducible source-only archive; models, environments and secrets stay local.
scripts/plot_release_results.py	Plot frozen aggregate proxies; these figures do not imply reviewed accuracy.
scripts/prepare_artifacts.py	Copy audited router metadata and attach existing adapters without changing originals.
scripts/prepare_safety_preview.py	Create a separate, versioned research bundle without promoting candidates.
scripts/probe_native_cache.py	Investigate BF16 cache differences against FP32 and multi-token greedy checks.
scripts/qualify_gptoss_candidate.py	Allocated-GPU validation, cache and real-train-row resource probe for one alternative.
scripts/qualify_gptoss_engine.py	Private original-MXFP4 vLLM qualification; BF16 tuned export remains a later gate.
scripts/qualify_gptoss_expert_export.py	Qualify a real Stage 3 checkpoint as a standard node-local HF model.
scripts/qualify_native.py	Allocated-GPU native BF16/cache/real-answer qualification, never a toy model.
scripts/qualify_native_precision.py	Bind raw cache diagnostics to an explicit, precision-aware adaptation gate.
scripts/reconcile_native_deployment.py	Correct an expired deployment record from Slurm evidence; never stop processes.
scripts/record_training_launch.py	Capture exact executed recipes/source/data hashes before GPU optimization.
scripts/recover_large_router.py	Strictly recover per-tensor chunks into a NEW standard checkpoint directory.
scripts/render_comprehensive_report.py	Render the full Markdown report as standalone HTML (Markdown==3.8.2).
scripts/render_research_report.py	Build a standalone HTML report from public Markdown and aggregate figures.
scripts/restart_safety_preview.py	Restart the private ICE research demo through an authenticated SSH master.
scripts/retire_archived_failed_export.py	One-time retirement of only the four restored, failed OLMoE generated shards.
scripts/start.sh	Shell environment/startup utility
scripts/start_native_gateway.py	Start the authenticated native gateway; never print credentials.
scripts/summarize_gptoss_completion.py	Verify frozen comparisons and generate an honest research completion report.
scripts/sync_native_source.py	Backed-up source rollout plus additive native namespaces; preserve external assets.
scripts/verify_bundle.py	Definitions:
scripts/verify_external_ice.py	Bind the frozen external release to actual ICE paths without moving weights.

Path	Responsibility / definitions
scripts/verify_handoff.py	Verify an extracted private continuation bundle using only Python stdlib.
scripts/verify_native_download.py	Verify pinned upstream LFS SHA-256, sizes, and safetensors headers.

configs — 13 files

Path	Responsibility / definitions
configs/internal/gptoss-experts-v2-native-eval.json	Configuration or deployment/template asset
configs/internal/gptoss-experts-v2.json	Configuration or deployment/template asset
configs/internal/gptoss-training-constraints.txt	Configuration or deployment/template asset
configs/internal/gptoss-v1-lr2e-6.json	Configuration or deployment/template asset
configs/internal/gptoss-v1.json	Configuration or deployment/template asset
configs/internal/native-v1.json	Configuration or deployment/template asset
configs/internal/native_safety_supplement.json	Configuration or deployment/template asset
configs/model_manifest.json	Configuration or deployment/template asset
configs/model_manifest_safety_preview.json	Configuration or deployment/template asset
configs/native_sources.json	Configuration or deployment/template asset
configs/runtime.example.json	Configuration or deployment/template asset
configs/safety_sources.json	Configuration or deployment/template asset
configs/safety_sources_v2.json	Configuration or deployment/template asset

tests — 39 files

Path	Responsibility / definitions
tests/test_api.py	Tests: ServiceDouble, settings, test_restricted_chat_and_compatibility, test_full_queue_is_retryable
tests/test_backend.py	Tests: test_adapter_selection_and_generation_are_serialized, test_queue_timeout_releases_capacity, test_checkpoint_label_order_is_explicit
tests/test_carry_native_reviews.py	Tests: records, test_exact_reuse_preserves_original_attribution_and_marks_not_independent, test_any_changed_assessment_input_prevents_reuse, test_stale_source_review_rejected
tests/test_cpu_demo.py	Tests: test_cpu_demo_is_authenticated_and_explicitly_simulated
tests/test_data.py	Tests: test_no_split_overlap_and_conflicts_quarantined
tests/test_expert_confirmatory_suite.py	Tests: test_selection_excludes_normalized_prior_and_cross_group_duplicates, test_selection_is_order_independent
tests/test_expert_pilot_controller.py	Controller proves readiness gates and crash-safe bounded scheduler behavior.
tests/test_generated_export_archive.py	Tests: GeneratedArchiveTest
tests/test_gptoss_calibration.py	A calibrated adapter retains a real learned update and correct projection algebra.

Path	Responsibility / definitions
tests/test_gptoss_completion_report.py	Tests: test_completed_collection_requires_matched_inputs_and_selected_checkpoint
tests/test_gptoss_ddp_qualification.py	Check actual DDP averaging for unequal response lengths before GPU qualification.
tests/test_gptoss_development.py	Ensure matched development controls and frozen-date export reuse.
tests/test_gptoss_distributed_resume.py	Real CPU Trainer transition: unchanged global batch, complete rank RNG, safe resume.
tests/test_gptoss_expert_adapter.py	Real miniature GPT-OSS architecture: native parity, training and standard export.
tests/test_gptoss_expert_export.py	Export gates detect numeric divergence and verify native cached decoding.
tests/test_gptoss_expert_training.py	Checkpoint correctness and deterministic sample cursor, independent of GPUs.
tests/test_gptoss_finish.py	Development selection must not turn failed exports into accepted candidates.
tests/test_gptoss_launcher_signal.py	Tests: test_checkpoint_signal_reaches_only_launchers_direct_worker
tests/test_gptoss_phase_archival.py	Tests: test_interim_answers_status_and_adapter_are_preserved_before_resume
tests/test_gptoss_selection_date.py	Tests: test_date_is_explicit_in_memory_and_original_template_value_unchanged, test_invalid_dates_are_rejected
tests/test_gptoss_training.py	Protect final-only supervision/output and validation quality selection.
tests/test_gptoss_validation_review.py	Reject comparisons with changed questions/answers instead of certifying proxies.
tests/test_internal_moe.py	Tests: test_upcycling_preserves_dense_output_and_dispatch, test_gate_and_selected_experts_receive_gradients, test_decoder_cannot_read_future_tokens, test_padding_excluded_from_router_lo
tests/test_native_accuracy.py	Incomplete or uncertain assessments must not become exact passing accuracy.
tests/test_native_data.py	Tests: row, test_protected_holdout_excludes_connected_training_context, test_publisher_conflicts_quarantine_whole_conversation, test_targets_conflict_and_duplicates_are_not_silently_sel
tests/test_native_export.py	Regression for the real OLMoE buffer mismatch invisible to weight hashes.
tests/test_native_load_format.py	Operational input counts must match the actual gateway template and reservation.
tests/test_native_model_archive.py	Tests: NativeArchiveTest
tests/test_native_operator.py	Exercise child startup ordering, readiness failures and clean signal exit.
tests/test_native_quota.py	Tests: test_overquota_marker_is_not_mistaken_for_available_filesystem_space, test_both_storage_and_file_limits_are_required, test_failed_checkpoint_is_quarantined_without_touching_previ
tests/test_native_service.py	Tests: Tokenizer, native_settings, transport, test_native_auth_contract_history_and_unavailable_profiles

Path	Responsibility / definitions
tests/test_native_submission_env.py	Tests: test_new_job_receives_its_own_gpu_rank_and_slurm_mapping
tests/test_operational_checkpoint.py	Wall-clock saves must preserve the Trainer's existing evaluation decisions.
tests/test_router_checkpointing.py	Real DeBERTa backward regression, run in the recovered training environment.
tests/test_safety.py	Tests: test_safety_has_priority_over_requested_math_and_base, test_benign_manual_override_and_legacy_router_compatibility, test_four_expert_router_cannot_start_without_safety_weights, t
tests/test_safety_improvements.py	Tests: test_calibrated_threshold_is_not_bypassed_by_top_safety_class, test_late_safety_evidence_reduces_group_loss_and_receives_gradient, test_benign_long_prompt_penalizes_safety_in_any
tests/test_safety_review.py	Tests: test_unreviewed_responses_do_not_receive_invented_zero_harm_scores, test_benign_source_unsafe_labels_do_not_dilute_reviewed_harmful_compliance
tests/test_source_release.py	Tests: packager, test_source_archive_excludes_private_and_generated_files, test_source_archive_rejects_embedded_credentials
tests/test_validation_answers.py	Tests: test_explicit_answer_is_not_replaced_by_later_context_numbers, test_does_not_accept_reference_mentions_or_earlier_bold_intermediate_values

deploy — 3 files

Path	Responsibility / definitions
deploy/Dockerfile	Configuration or deployment/template asset
deploy/aws/template.yaml	Configuration or deployment/template asset
deploy/compose.yaml	Configuration or deployment/template asset

Appendix D. Evidence catalog and original identities

References [E01]–[E40] identify the preserved records actually read for this report. Paths are relative to the research workspace unless prefixed `external-workspace/`, which denotes the preserved external application checkout. Raw artifacts may remain private or access-restricted; hashes are not a claim that those files ship with public source. The PDF/HTML is self-contained for interpretation of its results.

ID	Record	Preserved source path	SHA256 prefix
E01	External preservation inventory	`audit/releases/external-v1-ice.json`	`f573b81c8c453e79`
E02	Latest external/source preservation verification	`audit/completion-batch-20261021/preserved-external-artifacts-verified-20261022.json`	`8a3637a46d4cd981`
E03	External safety extension, v3 routing and comparison ledger	`external-workspace/audit/safety_improvement_v2.json`	`70030d2063d2ecfa`
E04	Native prepared-data manifest	`data/native/prepared/v1/manifest.json`	`b0c4d9671b68d013`
E05	Rejected OLMoE candidate outcome	`audit/native/olmoe-v1-candidate-outcome.json`	`f3d145f8c2a977e0`
E06	One/two-GPU numerical/scaling comparison	`outputs/internal/native/gptoss/ddp-qualification-6329565/comparison.json`	`cc7af7e1a36a09ab`
E07	Stage 2 archive restoration	`artifacts/internal/native/gptoss/v1/archive-20261018-r4-scored-calibrated/archive-verification.json`	`a2152576c1982abf`
E08	Four-trial calibration and selected Stage 3 archive	`audit/completion-batch-20261021/verified-calibration-results.json`	`0dc13a23d682b7b3`
E09	Real native expert qualification	`outputs/internal/native/gptoss/qualify-20261020-native-eval-v2/report.json`	`ddc4902404b7c41e`
E10	Frozen completion plan and selection identity	`audit/completion-batch-20261021/plan.json`	`89f5371a54f4dc1a`
E11	Expert-only routing telemetry	`outputs/internal/native/gptoss/moe-e2-f36368c989-expert-only/expert-usage.json`	`5a812ca707297341`
E12	Joint-gate routing telemetry	`outputs/internal/native/gptoss/moe-e2-f36368c989-joint-gates/expert-usage.json`	`a776d1d6dbdbc8ad`
E13	Earlier 526-answer attributed assessment	`audit/native/gptoss-r4-answer-assessment-20261019.json`	`b2d5491c2e363b66`
E14	Final 1,500-answer verified comparison	`outputs/internal/evaluation/finish-20261021/FINAL_RESEARCH_RESULTS.json`	`30b45b992b1aff0a`
E15	Post-hoc qualitative sample manifest	`audit/completion-batch-20261021/qualitative-review-sample.json`	`96882d472944c2d1`
E16	Historical passing cold start	`audit/native/archive-cold-start-6403909.json`	`251dae5abd893205`
E17	Historical full Stage 2 soak	`audit/native/load-soak-6403909/summary.json`	`d4b08c1df5745b34`
E18	Historical native restart and external rollback	`audit/native/recovery-6403909.json`	`5d9ed3c66e25ffb3`

ID	Record	Preserved source path	SHA256 prefix
E19	Latest real engine/gateway checks	audit/native/live-6479705.json	5d70597418eb9196
E20	Latest real cancellation/history/overload checks	audit/native/live-6479705-operations.json	c27a422ea556d370
E21	Latest desktop/mobile browser checks	audit/native/browser-6479705/report.json	c2fb6e30ea24e2e1
E22	Latest failed cold-start gate	audit/native/archive-cold-start-6479705.json	ceff61a9387ce8e7
E23	Latest stopped deployment record	audit/native/deployment-6479705.json	179a3880a70c15a5
E24	Home/scratch cleanup verification	audit/completion-batch-20261021/final-storage-verification-20261022.json	d8530453de1434ff
E25	Storage relocation and cleanup narrative	docs/STORAGE_CLEANUP_2026-10-21.md	26a1a2db1eb672f1
E26	Release-tree CPU test results	audit/release-20261022/release-tests.xml	1dbd9c05bd69d7e9
E27	Fresh local clone quick-start test results	audit/release-20261022/fresh-clone-tests.xml	dde0a0228c21ecd9
E28	Preserved GPT-OSS inference overlay inventory	audit/completion-batch-20261021/inference-environment-supplement-20261022.json	48691ba5a9ddeec4
E29	Parent qualitative review	audit/completion-batch-20261021/parent-qualitative-review-20261022.json	649c7d25aa391b4c
E30	Candidate qualitative review	audit/completion-batch-20261021/candidate-qualitative-review-20261022.json	594439dec22616e5
E31	Original immutable final research report	docs/FINAL_RESEARCH_REPORT_2026-10-21.md	b1083c842165f231
E32	Expert-only completed pilot	outputs/internal/native/gptoss/moe-e2-f36368c989-expert-only/report.json	c7c36295a901b65a
E33	Joint-gate completed pilot	outputs/internal/native/gptoss/moe-e2-f36368c989-joint-gates/report.json	a549d626df18b84d
E34	Stage 2 training and calibrated provenance	outputs/internal/native/gptoss/20261018-r4-scored-calibrated/sft/provenance.json	fb5311913888b081
E35	Final independent artifact audit	audit/completion-batch-20261021/FINAL_COMPLETION_AUDIT_20261022.json	e75342fbf2ab2b88
E36	Actual joint-gate pilot configuration and environment	outputs/internal/native/gptoss/moe-e2-f36368c989-joint-gates/metadata-6429777.json	d1f8d99053a98bea
E37	Pinned native data sources	configs/native_sources.json	763dd022d17b6569
E38	Pinned original safety data sources	configs/safety_sources.json	17a1c66abe7f04c3
E39	Pinned support/toxicity extension sources	configs/safety_sources_v2.json	5660130ef2b4d3aa
E40	Historical external safety experiment report	docs/SAFETY_RESULTS.md	acf20f188979120a

Full 64-character digests and byte counts are in `evidence-catalog.json` beside the rendered report. `verified-report-metrics.json` contains the checked answer hashes, cold-start outcomes and descriptive telemetry summaries. DOT/SVG/PNG figure sources are supplied in `figures/`; diagrams represent implemented paths and declared gates, not unsupported product capabilities.

Appendix E. Reproduction and reporting boundaries

The source CPU quick start is `python -m pip install -e '.[dev]'` followed by the targeted API/demo/data/packaging tests. The release also provides an exact CPU-demo dependency lock. Real GPU inference requires the matching upstream base, compatible tokenizer/adapters, separate model access, a qualified environment and an actual allocation. Full past research requires identity-matched prepared data and archived artifacts; a source-only clone does not supply them.

Read-only status verification uses `squeue`, `sacct`, exact job-bound reports and artifact checksums. Do not execute historical training/export commands merely because they are documented: experimentation is frozen. To rebuild this report, run `scripts/build_comprehensive_diagrams.py` and `scripts/complete_comprehensive_report.py`, then render the Markdown with the report renderer. Rebuilding uses saved evidence and creates no model experiment.

Some underlying metrics were generated by preexisting imperfect automatic scorers; the report preserves their limitations and versions. It does not fabricate human labels, silently relax thresholds, merge incompatible cohorts or convert a clean scheduler exit into a quality pass. Updated publication/deployment evidence should be appended with a new verification timestamp, preserving prior failed records.

Appendix F. Documentation and repository entry-point index

Older documentation is retained for provenance and may contain historical queue states or superseded plans. This comprehensive report and job-bound artifacts govern current status. Original reports are not silently rewritten to hide failed experiments. The following index locates the rest of the project documentation.

Path	Document / entry point
docs/ARCHITECTURE.md	Architecture
docs/DEPLOYMENT.md	Restricted deployment with zero AWS spend
docs/FINAL_RESEARCH_REPORT_2026-10-21.md	Stage 2–3 final research report
docs/GPTOSS_RECOVERY_2026-10-18.md	GPT-OSS regression recovery — October 18, 2026
docs/INTERNAL_MOE.md	Internal MoE development
docs/INTERNAL_MOE_PRODUCT_PROPOSAL.md	External-to-internal MoE: deployable product proposal
docs/INTERNAL_PRODUCT_RUNBOOK.md	Native internal MoE on ICE
docs/MOE_DUAL_TRACK_ROADMAP.md	External and internal MoE: staged product roadmap
docs/NATIVE_GPTOSS_EXECUTION.md	Native alternative execution — October 17, 2026
docs/NATIVE_REVIEW_RUBRIC.md	Native v1 restricted research answer review
docs/OPERATIONS.md	ICE operation
docs/PROJECT_RELEASE_PLAN_2026-10-22.md	Project completion, deployment and publication plan
docs/QUALITY.md	Model quality evidence
docs/RELEASE_PROGRESS_2026-10-22.md	Release progress — October 22, 2026
docs/REMOTE_SESSION_HANDOFF.md	ICE continuation handoff
docs/RESEARCH_RELEASE.md	Restricted research release
docs/RESULTS.md	Verified results and release scope
docs/SAFETY_EXPERT.md	Fourth expert: safety alignment
docs/SAFETY_IMPROVEMENT_V2.md	Safety improvement experiment v2
docs/SAFETY_RESULTS.md	Safety experiment results
docs/SAFETY_REVIEW_GUIDE.md	Next steps for safety quality
docs/STAGE23_COMPLETION_2026-10-21.md	Stage 2–3 bounded completion batch — October 21, 2026
docs/STAGE23_RUNS_2026-10-20.md	Stage2 and Stage3 queued experiments — October20,2026
docs/STAGE2_ACCEPTANCE_2026-10-19.md	Stage 2 acceptance evidence, October 19, 2026
docs/STATUS.md	Implementation status
docs/STORAGE_CLEANUP_2026-10-21.md	Home quota recovery — October 21, 2026
docs/models/SAFETY_LORA_V2.md	SAFETY LoRA v2 model card
docs/models/SAFETY_LORA_V3.md	SAFETY LoRA v3 model card

Path	Document / entry point
docs/models/SAFETY_ROUTER_V3.md	SAFETY router v3 model card
README.md	External MoE Router
README_STAGE_1_2.md	Dual-Profile Mixture-of-Experts: Recovery, Adaptation, and Serving
AGENTS.md	Georgia Tech ICE / Slurm Research Compute Instructions
PROJECT_COMPLETION_PLAN.md	External MoE recovery and completion plan
LICENSE	LICENSE
pyproject.toml	pyproject.toml
requirements-api.lock	requirements-api.lock
.gitignore	.gitignore
.env.example	Artifact root must contain base/, router/, math_expert/, science_expert/, general_expert/.
.github/workflows/checks.yml	checks.yml

Appendix G. Requirement-to-evidence traceability

Requirement	Implementation/evidence	Current decision
Preserve existing specialists	Frozen weight inventory and original source archive, E01–E02	Verified
Correct request routing	Named adapter selection, checkpoint label metadata and lock	Implemented and integrated
Add support/safety coverage	Published-data extension and v3 ledger, E03	Integration complete; routing gates failed
Avoid fabricated data/reviews	Source manifests, review importer and blank masked forms	No synthetic native training examples; human review pending
Prevent exact project split leakage	Grouped source preparation and protected keys, E04	Checks passed; semantic contamination unknown
Train native model reproducibly	Provenance/config/checkpoint identity, E34	Stage 2 completed; quality failed
Justify multi-GPU work	Fixed-batch numerical/scaling comparison, E06	Qualified only within measured benchmark
Export actual learned updates	Projection/frozen-state/reload probes and archives, E07–E10	Selected research exports qualified
Train actual FFN experts/gates	Qualified resume, changed-state and pilot reports, E09/E32/E33	Both pilots completed
Compare selected model fairly	Frozen protocol, 500 paired input hashes, E14	No demonstrated held-out proxy gain
Operate private app reliably	API/browser/load/recovery, E16–E23	Historical full pass; latest cold start failed
Keep resources/storage bounded	QoS policy, scratch helper, cleanup proof, E24–E25	Implemented policy; quotas require ongoing checks
Provide reproducible public source	Export manifests and CPU/fresh-clone checks, E26–E27	Local candidate prepared; remote publication pending
Establish independent quality	Masked review sheet and reviewer protocol	Pending; not replaced by AI assessments
Accept production deployment	All relevant integrity, operations, quality and review gates	False; research scope only

Appendix H. Full declared configurations and pinned source registry

These are the saved configuration objects, not newly chosen hyperparameters. Stage 2 provenance also records its post-training calibration. Stage 3's runtime variant flag chooses expert-only versus joint gates while sharing the configuration below. Paths refer to the prepared research workspace, whose required artifacts are not included by a source-only clone. [E34, E36]

Stage 2 recorded configuration


```
{
  "profile": "native-internal-v1",
  "candidate_release": "native-internal-v1-gptoss-r4-scored-calibrated",
  "repository": "openai/gpt-oss-20b",
  "revision": "6cee5e81ee83917806bbde320786a8fb61efebee",
  "base": "artifacts/internal/native/candidates/gpt-oss-20b/base",
  "data": "data/native/prepared/v1",
  "max_length": 1024,
  "context_limit": 2048,
  "output_limit": 256,
  "rank": 8,
  "alpha": 16,
  "target_modules": [
    "q_proj",
    "k_proj",
    "v_proj",
    "o_proj"
  ],
  "experts_and_gates_frozen": true,
  "epochs": 1,
  "gradient_accumulation": 16,
  "learning_rate": 2e-06,
  "eval_steps": 300,
  "answer_validation_per_source": 64,
  "seed": 42,
  "production_accepted": false,
  "access": "restricted_noncommercial_research"
}
```


Stage 3 recorded configuration


```
{
  "profile": "native-internal-v2-expert-pilot",
  "parent_release": "native-internal-v1-gptoss-r4-scored-calibrated",
  "archive": "artifacts/internal/native/gptoss/v1/archive-20261018-r4-scored-calibrated",
  "tokenizer": "artifacts/internal/native/candidates/gpt-oss-20b/base",
  "data": "data/native/prepared/v1",
  "rank": 2,
  "alpha": 2.0,
  "max_length": 512,
  "train_per_source": 200,
  "validation_per_source": 16,
  "effective_batch": 2,
```

```
  "max_steps": 200,
  "expert_learning_rate": 1e-05,
  "gate_learning_rate": 1e-06,
  "gate_warmup_steps": 20,
  "qualification_gate_warmup_steps": 2,
  "eval_steps": 50,
  "checkpoint_seconds": 1200,
  "seed": 42,
  "production_accepted": false,
  "answer_quality_accepted": false,
  "access": "restricted_noncommercial_research",
  "question": "Can actual FFN low-rank updates followed by native gate updates improve fixed validation response NLL
over the preserved Stage 2 candidate without changing its architecture?",
  "scope": "Qualification and bounded pilot only; independent generated-answer and merge/reload checks required for
Stage 3 acceptance",
  "template_date": "2026-10-19",
  "change_note": "Preserve original tokenized date while correcting native CUDA evaluation arithmetic; datasets, seeds,
rank and optimizer settings unchanged."
}
```


Actual pilot environment

The joint-gate metadata records CUDA 12.8 and the following package versions. A CUDA version recorded by Torch is not a claim that every system driver/toolkit component has the same version.

Package	Recorded version
torch	2.8.0
transformers	4.57.5
peft	0.18.1

Pinned source identities

These source registries distinguish upstream revision identity from a locally prepared split hash. Recorded terms are provenance, not blanket redistribution permission. The source files and inherited manifests must also pass integrity checks before reproducing the original preparation. [E37–E39]

Registry / source	Repository or provider	Pinned revision	Recorded terms
native_sources.json / gsm8k	openai/grade-school-math	`3101c7d5072418e28b9008a6636bde82a006892c`	MIT
native_sources.json / sciq	allenai/sciq	`2c94ad3e1aafab77146f384e23536f97a4849815`	CC-BY-NC-3.0
native_sources.json / dolly	databricks/databricks-dolly-15k	`bdd27f4d94b9c1f951818a7da7fd7aeea5dbff1a`	CC-BY-SA-3.0
safety_sources.json / aegis	nvidia/Aegis-AI-Content-Safety-Dataset-2.0	`d86bb8bedff51d25ac834ab7838f1cc61acb7a2c`	CC-BY-4.0
safety_sources.json / pku	PKU-Alignment/PKU-SafeRLHF	`9421ffafec3fa40a1f1a7d567b4d525079477ecb`	CC-BY-NC-4.0
safety_sources.json / do_not_answer	LibrAI/do-not-answer	`74e74f2e4507ef256fe536f78a776f4a1ff67955`	Apache-2.0

Registry / source	Repository or provider	Pinned revision	Recorded terms
safety_sources_v2.json / esconv	thu-coai/Emotional-Support-Conversation	`f262d062ad74cb39b17ea476facc81568ddcba24`	Academic research only; copyright CoAI Group, Tsinghua University
safety_sources_v2.json / toxicchat	lmsys/toxic-chat	`29df8e4dba60e1f4af4b4075c0705c5b313548a8`	CC-BY-NC-4.0